

BONY ORBIT

The bony orbits are skeletal cavities located on either side of the root of the nose that serve as sockets for the eyes and associated tissues. The walls of each orbit protect the eye from injury, provide points of attachment for six extraocular muscles that allow the accurate positioning of the visual axis, and determine the spatial relationship between the two eyes, which is essential for both binocular vision and conjugate eye movements.

By convention, each cavity is considered to approximate to a quadrilateral pyramid with its base at the orbital opening, narrowing to its apex along a posteromedially directed axis. Each orbit has a roof, floor and medial and lateral walls. The medial walls lie approximately 25 mm apart in adults and are nearly parallel. The angle between the medial and lateral walls is about 45°. The compromise between protection and ensuring a good field of view dictates that each eyeball is located anteriorly within the orbit. The eyeball thus occupies only one-fifth of the volume of the orbit (**Fig. 41.1**); the remainder of the cavity is filled with extraocular muscles, vessels and nerves that are contained within and supported by orbital fat and connective tissue. In brief, the orbit transmits the optic, oculomotor, trochlear and abducens nerves, and branches of the ophthalmic and maxillary divisions of the trigeminal nerve, the ciliary parasympathetic ganglion and the ophthalmic vessels. It also contains the nasolacrimal apparatus that mediates tear drainage into the nasal cavity.

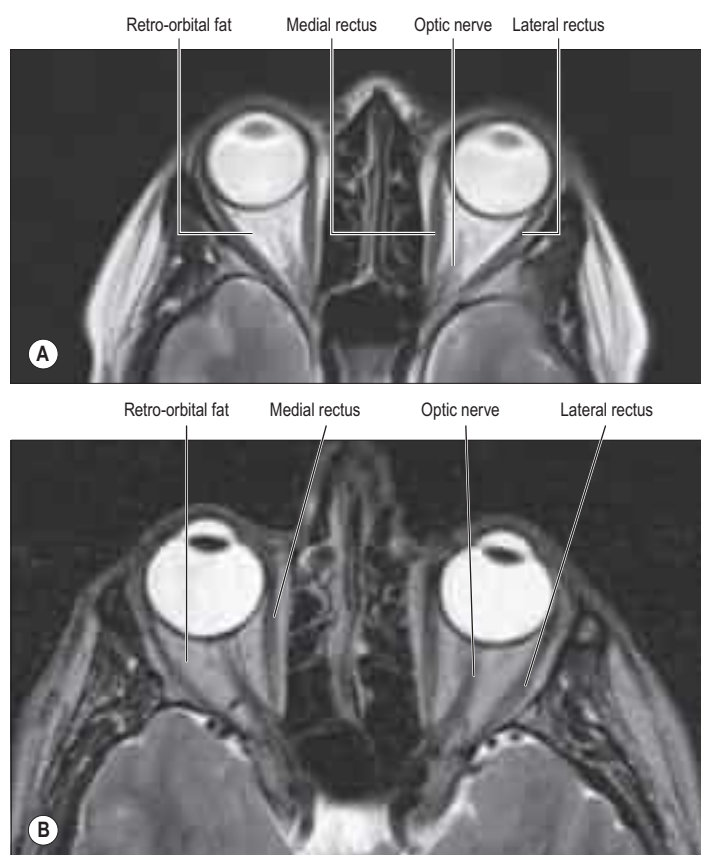


Fig. 41.1 **A**, A T2-weighted axial magnetic resonance imaging (MRI) scan through the mid-orbit. **B**, Note optic nerve in its entirety. (Courtesy of Dr Timothy Beale FRCS, Royal National Throat Nose and Ear Hospital, London.)

There is a strong correlation between the pattern of orbital growth and eyeball growth. Orbital growth is most rapid during the first 12–24 months of life and most parameters reach 86–96% of adult values by the age of 8 years (**Escaravage and Dutton 2013**). Orbital volumes are larger in boys than in girls throughout childhood (**Bentley et al 2002**).

Roof

The roof of the orbit is formed principally by the thin orbital plate of the frontal bone (**Fig. 41.2**). It is gently concave on its orbital aspect, which separates the orbital contents and the brain in the anterior cranial fossa. Anteromedially, it contains the frontal sinus and displays a small trochlear fovea, sometimes surmounted by a small spine, where the cartilaginous trochlea (pulley) for superior oblique is attached. Anterolaterally, there is a shallow fossa that houses the orbital part of the lacrimal gland. The roof slopes down significantly towards the apex, joining the lesser wing of the sphenoid, which completes the roof. The optic canal lies between the roots of the lesser wing and is bounded medially by the body of the sphenoid.

Medial wall

The medial wall of the orbit is formed principally by the orbital plate (lamina papyracea) of the ethmoid bone (see **Fig. 41.2**). This paper-thin, rectangular plate covers the middle and posterior ethmoidal air cells, providing a route by which infection can spread into the orbit. The ethmoid articulates with the medial edge of the orbital plate of the frontal bone at a suture that is interrupted by anterior and posterior ethmoidal foramina. Posteriorly, it articulates with the body of the sphenoid, which forms the medial wall of the orbit to its apex. The lacrimal bone lies anterior to the ethmoid; it contains a fossa for the nasolacrimal sac that is limited in front by the anterior lacrimal crest on the frontal process of the maxilla and behind by the posterior lacrimal crest of the lacrimal bone (to which the lacrimal part of orbicularis oculi and lacrimal fascia are attached). A descending process of the lacrimal bone at the lower end of the posterior lacrimal crest contributes to the formation of the upper part of the nasolacrimal canal, which is completed by the maxilla (**Fig. 41.3**). During development, the medial wall of the orbit doubles in length, with disproportionate enlargement of its anterior half. Growth is rapid during the first 6 years of life and gradual between 7 years and adulthood (**Isaacson and Monge 2003**).

Floor

The floor of the orbit is mostly formed by the orbital plate of the maxilla, which articulates with the zygomatic bone anterolaterally and the small triangular orbital process of the palatine bone posteromedially (see **Fig. 41.3**). The floor is thin and largely roofs the maxillary sinus. Not quite horizontal, it ascends a little laterally. Anteriorly, it curves into the lateral wall, and posteriorly, it is separated from the lateral wall by the inferior orbital fissure, which connects the orbit posteriorly to the pterygopalatine fossa, and more anteriorly to the infratemporal fossa. The medial lip is notched by the infraorbital groove. The latter passes forwards and sinks into the floor to become the infraorbital canal, which opens on the face at the infraorbital foramen; the infraorbital groove, canal and foramen contain the infraorbital nerve and vessels. Proportionally more pure orbital fractures involve the floor, particularly in the region of the infraorbital groove (**Burm et al 1999**). The classic 'blowout fracture' leaves the orbital rim intact and typically entraps soft tissue structures, leading to diplopia, impaired ocular motility and enophthalmos; infraorbital nerve involvement leads to ipsilateral sensory disturbance of the skin of the midface.

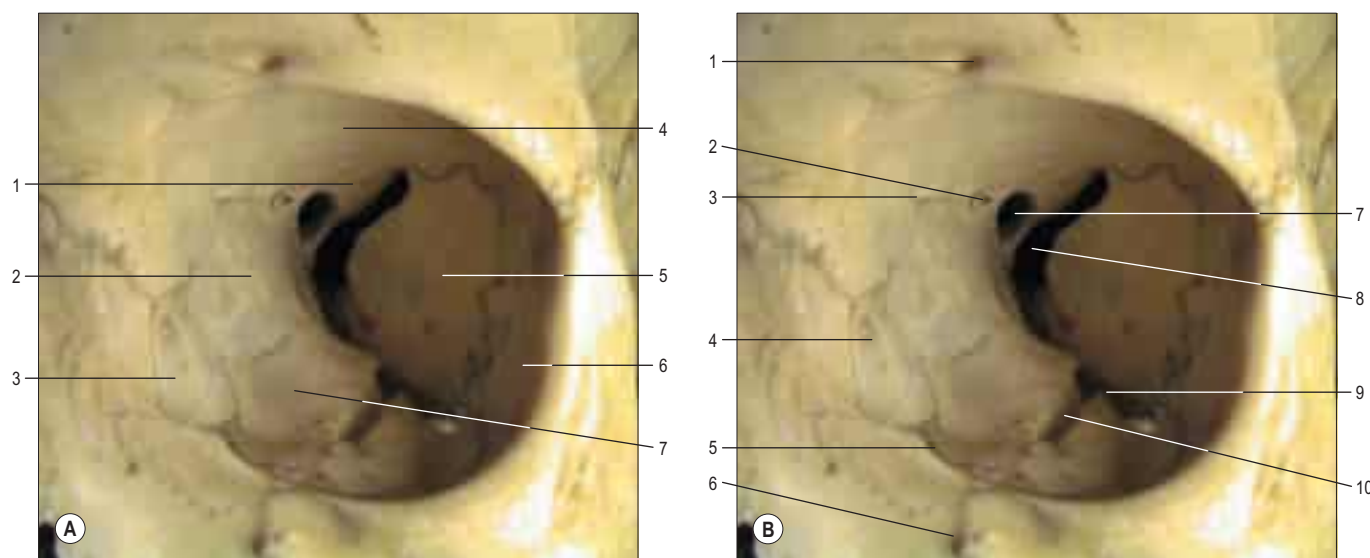


Fig. 41.2 An anterior view of the left orbit indicating the bones forming the walls (**A**) and the principal apertures (**B**). **A**, Key: 1, lesser wing of sphenoid; 2, orbital plate of ethmoid; 3, lacrimal bone; 4, orbital plate of frontal bone; 5, greater wing of sphenoid; 6, zygomatic; 7, orbital plate of maxilla. **B**, Key: 1, supraorbital foramen; 2, posterior ethmoidal foramen; 3, anterior ethmoidal foramen; 4, lacrimal fossa; 5, (canal for) nasolacrimal duct; 6, infraorbital foramen; 7, optic canal; 8, superior orbital fissure; 9, inferior orbital fissure; 10, infraorbital groove.

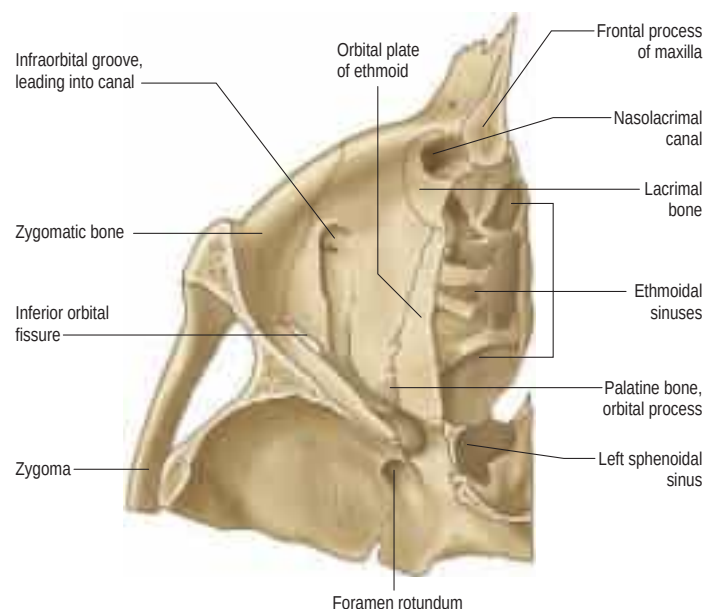


Fig. 41.3 A horizontal section through the floor of the left orbit and the adjacent ethmoidal and sphenoidal sinuses, viewed from above.

Lateral wall

The lateral wall of the orbit is formed by the orbital surface of the greater wing of the sphenoid posteriorly and the frontal process of the zygomatic bone anteriorly; the bones meet at the sphenozygomatic suture. The zygomatic surface contains the openings of minute canals for the zygomaticofacial and zygomaticotemporal nerves, the former near the junction of the floor and lateral wall, and the latter at a slightly higher level, sometimes near the suture. The orbital tubercle, to which the lateral palpebral ligament, the check ligament of lateral rectus and the aponeurosis of levator palpebrae are all attached, lies just inside the midpoint of the lateral orbital margin. The lateral wall is the thickest wall of the orbit, especially posteriorly, where it separates the orbit from the middle cranial fossa. Anteriorly, the lateral wall separates the orbit and the infratemporal fossa. The lateral wall and roof are continuous anteriorly but are separated posteriorly by the superior orbital fissure, which lies between the greater wing (below) and lesser wing (above) of the sphenoid, and communicates with the middle cranial fossa. The fissure tapers laterally but widens at its medial end, its long axis descending posteromedially. Where the fissure begins to widen, its inferolateral edge shows a projection, often a spine, for the lateral attachment of the common tendinous ring (see Fig. 41.4). An infraorbital sulcus, which runs from the superolateral end of the superior orbital fissure towards

the orbital floor, is sometimes associated with an anastomosis between the middle meningeal and infraorbital arteries.

ORBITAL FISSURES AND FORAMINA

Optic canal

The lesser wing of the sphenoid is connected to the body of the sphenoid by a thin, flat anterior root and a thick, triangular posterior root. The optic canal lies between these (see Fig. 41.2) and connects the orbit to the middle cranial fossa, transmitting the optic nerve and its meningeal sheaths, and the ophthalmic artery. The common tendinous ring, which gives origin to the four recti, is attached to the bone near the superior, medial and lower margins of the orbital opening of the canal (see Fig. 41.4). The diameter of the cranial opening of the optic canal increases significantly during the fetal period and childhood into adulthood.

Superior orbital fissure

The superior orbital fissure is the gap between the greater and lesser wings of the sphenoid, bounded medially by the body of the sphenoid, and closed at its anterior extremity by the frontal bone (see Fig. 41.2). It connects the cranial cavity with the orbit and transmits the oculomotor, trochlear and abducens nerves, branches of the ophthalmic nerve and the ophthalmic veins (see Fig. 41.4).

Inferior orbital fissure

The inferior orbital fissure is bounded above by the greater wing of the sphenoid, below by the maxilla and the orbital process of the palatine bone, and laterally by the zygomatic bone (see Fig. 41.2). The maxilla and sphenoid often meet at the anterior end of the fissure, excluding the zygomatic bone. The inferior orbital fissure connects the orbit with the pterygopalatine and infratemporal fossae and transmits the infraorbital and zygomatic branches of the maxillary nerve and accompanying vessels (see Fig. 41.4), orbital rami from the pterygopalatine ganglion and a connection between the inferior ophthalmic vein and pterygoid venous plexus. A small maxillary depression may mark the attachment of inferior oblique anteromedially, lateral to the lacrimal hamulus.

Ethmoidal foramina

The anterior and posterior ethmoidal foramina usually lie in the fronto-ethmoidal suture (see Fig. 41.2). The posterior foramen may be absent, and occasionally there is a middle ethmoidal foramen. The foramina open into canals that transmit their vessels and nerves into the ethmoidal sinuses, anterior cranial fossa and nasal cavity.

COMMON TENDINOUS RING

The common tendinous ring is a fibrous ring that surrounds the optic canal and part of the superior orbital fissure at the apex of the orbit, and gives origin to the four recti (Fig. 41.4). The optic nerve and ophthalmic artery enter the orbit via the optic canal, and so lie within the common tendinous ring. The superior and inferior divisions of the oculomotor nerve, the nasociliary branch of the ophthalmic nerve, and the abducens nerve also enter the orbit within the common tendinous ring, but they do so via the superior orbital fissure (see Fig. 41.15). The trochlear nerve and the frontal and lacrimal branches of the ophthalmic

nerve all enter the orbit through the superior orbital fissure but lie outside the common tendinous ring. Structures that enter the orbit through the inferior orbital fissure lie outside the common tendinous ring. The close anatomical relationship of the optic nerve and other cranial nerves at the orbital apex means that lesions in this region may lead to a combination of visual loss from optic neuropathy and ophthalmoplegia from multiple cranial nerve involvement (Yeh and Foroozan 2004).

ORBITAL CONNECTIVE TISSUE AND FAT

The orbit contains a complex arrangement of connective tissue that forms a supporting framework for the eyeball and also influences ocular rotations and compartmentalizes orbital fat (Fig. 41.5). Certain regions have anatomical and clinical significance, including the orbital septum, fascial sheath of the eye, 'check' ligaments, suspensory ligament and periosteum. The notion that orbital connective tissues function as extraocular muscle pulleys and influence ocular motility has recently gained widespread acceptance (Demer 2002, Miller 2007).

ORBITAL SEPTUM

The orbital septum is a weak membranous sheet, attached to the orbital rim where it becomes continuous with the periosteum (see Fig. 41.5). It extends into each eyelid and blends with the tarsal plates and, in the upper eyelid, with the superficial lamella of levator palpebrae superioris. The orbital septum is thickest laterally, where it lies in front of the lateral palpebral ligament. It passes behind the medial palpebral ligament and nasolacrimal sac, but in front of the pulley of superior oblique. The septum is pierced above by levator palpebrae superioris and below by a fibrous extension from the sheaths of inferior rectus and inferior oblique. The lacrimal, supratrochlear, infratrochlear and supraorbital nerves and vessels pass through the septum from the orbit *en route* to the face and scalp. Clinically, the septum is an important anatomical reference to differentiate pre- and postseptal (orbital) cellulitis.

FASCIAL SHEATH OF THE EYEBALL

A thin fascial sheath, the fascia bulbi (Tenon's capsule), envelops the eyeball from the optic nerve to the corneoscleral junction, separating it from the orbital fat and forming a socket for the eyeball (see Fig. 41.5; Fig. 41.6). The ocular aspect of the sheath is loosely attached to the sclera by delicate bands of episcleral connective tissue. Posteriorly, it is traversed by ciliary vessels and nerves. It fuses with the sclera and with the sheath of the optic nerve where the latter enters the eyeball; attachment to the sclera is strongest in this position and again anteriorly, just behind the corneoscleral junction at the limbus. Injection of local anaesthetics via a cannula into the space between the fascia bulbi and

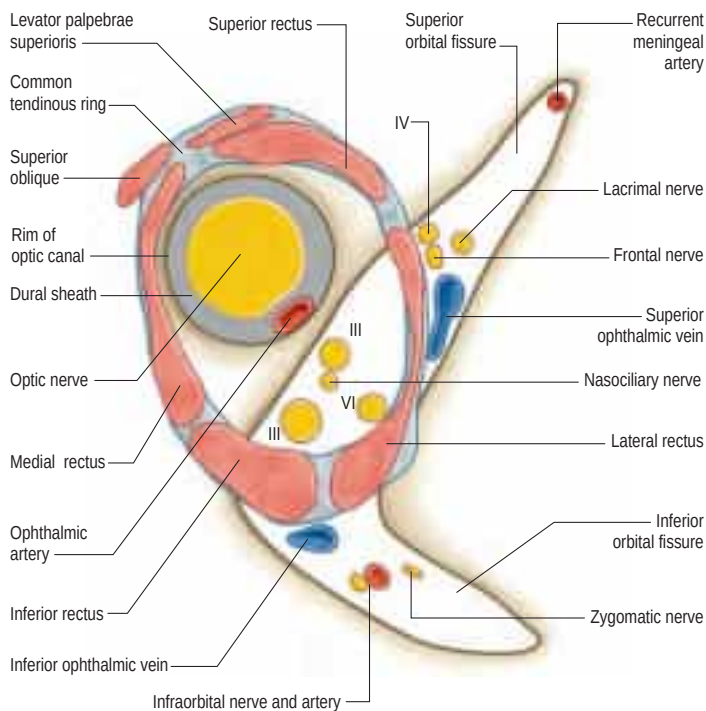


Fig. 41.4 The common tendinous ring: muscle origins have been superimposed. The relative positions of the nerves and vessels that enter the orbital cavity by passing through the superior orbital fissure or optic canal are shown. Note that the attachments of levator palpebrae superioris and superior oblique lie external to the common tendinous ring but are attached to it. The ophthalmic veins frequently pass through the ring. The recurrent meningeal artery (a branch of the ophthalmic artery) is often conducted from the orbit to the cranial cavity through its own foramen. Abbreviations: III, oculomotor nerve; IV, trochlear nerve; VI, abducens nerve.

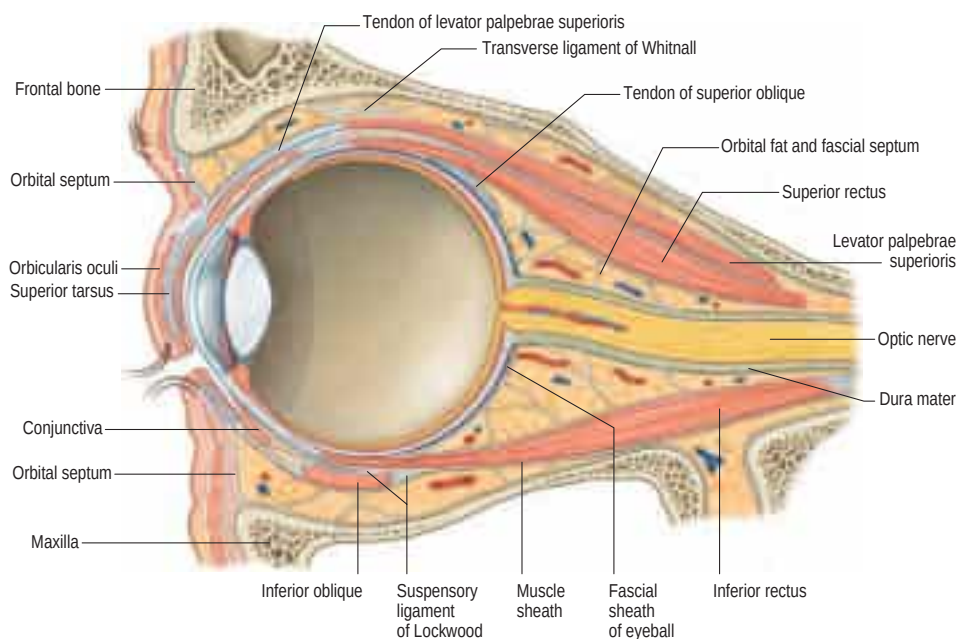


Fig. 41.5 The contents of the orbit, sagittal section.

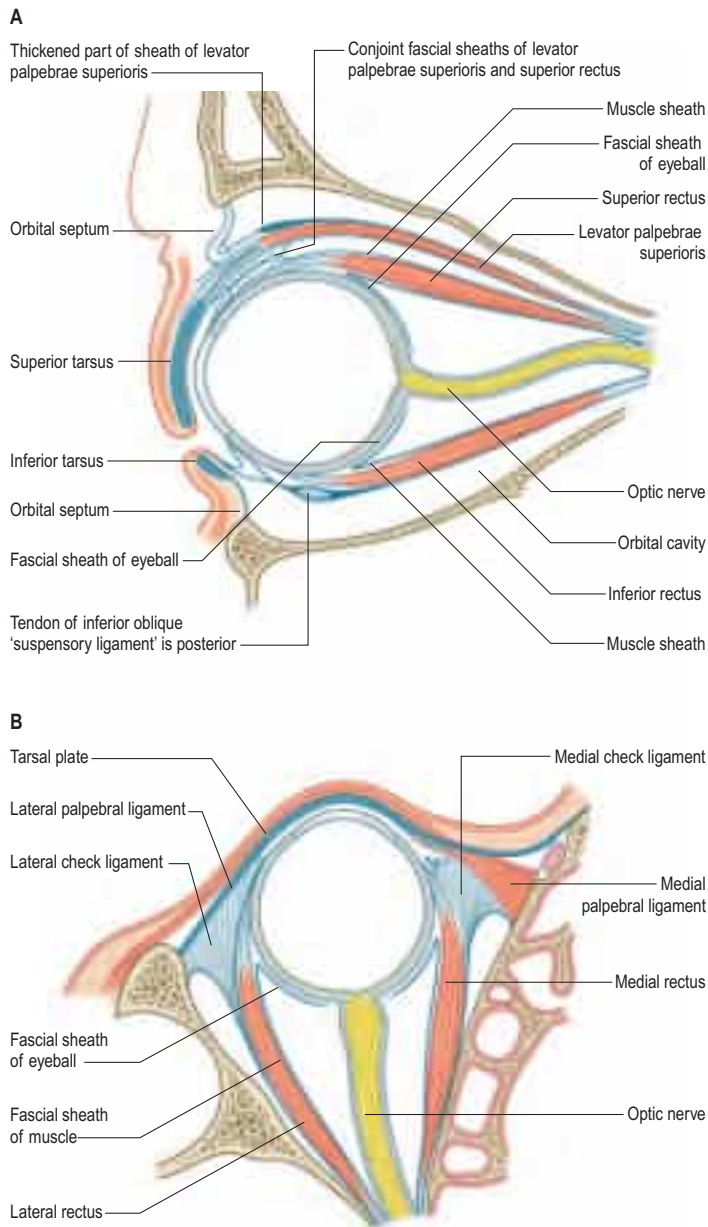


Fig. 41.6 **A**, The orbital fascia, sagittal section. **B**, The orbital fascia, horizontal section.

the sclera (sub-Tenon's anaesthesia) has become a popular technique for many ophthalmic surgical procedures (Davison et al 2007).

The fascia bulbi is perforated by the tendons of the extraocular muscles and is reflected on to each as a tubular sheath, the muscular fascia. The sheath of superior oblique reaches the fibrous pulley (trochlea) associated with the muscle. The sheaths of the four recti are very thick anteriorly but are reduced posteriorly to a delicate perimysium. Just before they blend with the fascia bulbi, the thick sheaths of adjacent recti become confluent and form a fascial ring.

Expansions from the muscular fascia are important for the attachments they make. Those from the medial and lateral recti are triangular and strong, and are attached to the lacrimal and zygomatic bones, respectively; since they may limit the actions of the two recti, they are termed the medial and lateral check ligaments (see Fig. 41.6). Other extraocular muscles have less substantial check ligaments, and the capacity of any of them actually to limit movement has been questioned.

The sheath of inferior rectus is thickened on its underside and blends with the sheath of inferior oblique. These two, in turn, are continuous with the fascial ring noted earlier and therefore with the sheaths of the medial and lateral recti. Since the latter are attached to the orbital walls by check ligaments, a continuous fascial band, the suspensory ligament of the eye, is slung like a hammock below the eye, providing sufficient support such that, even when the maxilla (forming the floor of the orbit) is removed, the eye will retain its position.

The thickened fused sheath of inferior rectus and inferior oblique also has an anterior expansion into the lower eyelid, where, augmented



Fig. 41.7 A coronal section through the left orbit (viewed from in front), cut through a plane passing 5 mm behind the posterior pole of the globe. Abbreviations: atc, adipose tissue compartments; eb, ethmoid bone; fn, frontal nerve; frb, frontal bone; ir, inferior rectus; lps, levator palpebrae superioris; lr, lateral rectus; m, maxilla (bone); mr, medial rectus; ncn, nasociliary nerve; opn, optic nerve; som, superior oblique; sov, superior ophthalmic vein; sr, superior rectus. (With permission from Koornneef L 1977 The architecture of the musculo-fibrous apparatus in the human orbit. *Acta Morphol Neerl Scand* 15:35–64.)

by some fibres of orbicularis oculi, it attaches to the inferior tarsus as the inferior tarsal muscle; contraction of inferior rectus in downward gaze therefore also draws the lid downward. The sheath of levator palpebrae superioris is also thickened anteriorly, and just behind the aponeurosis it fuses inferiorly with the sheath of superior rectus. It extends forwards between the two muscles and attaches to the upper fornix of the conjunctiva.

Other extensions of the fascia bulbi pass medially and laterally, and attach to the orbital walls, forming the transverse ligament of the eye. This structure is of uncertain significance, but presumably plays a part in drawing the fornix upwards in gaze elevation and may act as a fulcrum for levator movements. Other numerous finer fasciae form radial septa that extend from the fascia bulbi and the muscle sheaths to the periosteum of the orbit, and so provide compartments for orbital fat. They also prevent the gross displacement of orbital fat, which could interfere with the accurate positioning of the two eyes that is essential for binocular vision.

The periosteum of the orbit is only loosely attached to bone. Behind, it is united with the dura mater surrounding the optic nerve and, in front, it is continuous with the periosteum of the orbital margin, where it gives off a stratum that contributes to the orbital septum. It also attaches to the trochlea and, as the lacrimal fascia, forms the roof and lateral wall of the fossa for the nasolacrimal sac.

Orbital connective tissue pulleys

There is mounting evidence that challenges the traditional view that the recti are attached only at their origin and scleral insertion. The concept that orbital connective tissue sheaths elastically coupled to the orbital walls function as pulleys was initially proposed as an explanation for the observed orbital stability of rectus muscle paths (Miller 1989). Each pulley consists of an encircling sleeve of collagen located within the fascia bulbi, near the equator of the globe. Elastic fibres and bundles of smooth muscle confer the required internal rigidity to the structure (Demer 2002). Although the original model described a passive pulley system, the current view is that fibres from the orbital surface of the muscle insert into the pulley sleeve to allow small longitudinal movements. This 'active pulley hypothesis' provides a better explanation for normal ocular kinematics (Miller 2007).

ORBITAL FAT

The spaces between the main structures of the orbit are occupied by fat, particularly in the region between the optic nerve and the surrounding cone of muscles (see Fig. 41.5; Fig. 41.7). Fat also lies between the muscles and periosteum, and is limited anteriorly by the orbital septum. Collectively, the fat helps to stabilize the position of the eyeball and also acts as a socket within which the eye can rotate. Conditions

resulting in an increased overall volume of orbital fat with associated swelling of the extraocular muscles, e.g. hyperthyroidism (Graves' disease), may lead to forward protrusion of the eyeball (exophthalmos).

EXTRAOCULAR MUSCLES

There are seven skeletal extraocular (extrinsic) muscles associated with the eye. Levator palpebrae superioris is an elevator of the upper eyelid, and the other six, i.e. four recti (superior, inferior, medial and lateral) and two obliques (superior and inferior), are capable of moving the eye in almost any direction. Complete congenital absence of the extraocular muscles, thought to represent a severe form of congenital fibrosis syndrome, has been described (Brady et al 1992). Rarely, humans have deep orbital bands consistent with supernumerary extraocular muscles (Khitri and Demer 2010).

LEVATOR PALPEBRAE SUPERIORIS

Levator palpebrae superioris is a thin, triangular muscle that arises from the inferior aspect of the lesser wing of the sphenoid, above and in front of the optic canal, and separated from it by the attachment of superior rectus (see Fig. 41.4). It has a short narrow tendon at its posterior attachment and broadens gradually, then more sharply as it passes anteriorly above the eyeball. The muscle ends in front in a wide aponeurosis. Some of its tendinous fibres pass straight into the upper eyelid to attach to the anterior surface of the tarsus, while the rest radiate and pierce orbicularis oculi to pass to the skin of the upper eyelid. A thin lamina of smooth muscle, the superior tarsal muscle, passes from the underside of levator palpebrae superioris to the upper margin of the superior tarsus (see Fig. 41.20).

The connective tissue sheaths of the adjoining surfaces of levator palpebrae superioris and superior rectus are fused (see Fig. 41.5). Where the two muscles separate to reach their anterior attachments, the fascia between them forms a thick mass to which the superior conjunctival fornix is attached; this is usually described as an additional attachment of levator palpebrae superioris. Traced laterally, the aponeurosis of the levator passes between the orbital and palpebral parts of the lacrimal gland to attach to the orbital tubercle of the zygomatic bone. Traced medially, it loses its tendinous nature as it passes closely over the reflected tendon of superior oblique, and continues on to the medial palpebral ligament as loose strands of connective tissue.

Vascular supply Levator palpebrae superioris receives its arterial supply both directly from the ophthalmic artery and indirectly from its supraorbital branch.

Innervation Levator palpebrae superioris is innervated by a branch of the superior division of the oculomotor nerve that enters the inferior surface of the muscle. Sympathetic fibres to the smooth muscle component of levator palpebrae superioris (superior tarsal muscle) are derived from the plexus surrounding the internal carotid artery; these nerve fibres may join the oculomotor nerve in the cavernous sinus and pass forwards in its superior branch.

Actions Levator palpebrae superioris elevates the upper eyelid. During this process, the lateral and medial parts of its aponeurosis are stretched and thus limit its action; the elevation is also checked by the orbital septum. Elevation of the eyelid is opposed by the palpebral part of orbicularis oculi. Levator palpebrae superioris is linked to superior rectus by a check ligament; thus the upper eyelid elevates when the gaze of the eye is directed upwards.

The position of the eyelids depends on reciprocal tone in orbicularis oculi and levator palpebrae superioris, and on the degree of ocular protrusion. In the opened position, the upper eyelid covers the upper part of the cornea, while the lower lid lies just below its lower margin. The eyes are closed by movements of both lids, produced by the contraction of the palpebral part of orbicularis oculi and relaxation of levator palpebrae superioris. In looking upwards, the levator contracts and the upper lid follows the ocular movement. At the same time, the eyebrows are also usually raised by the frontal parts of occipitofrontalis to diminish their overhang. The lower lid lags behind ocular movement, so that more sclera is exposed below the cornea and the lid is bulged a little by the lower part of the elevated eye. When the eye is depressed, both lids move; the upper retains its normal relation to the eyeball and still covers about a quarter of the cornea, whereas the lower lid is depressed because the extension of the thickened fascia of inferior rectus and inferior oblique pull on its tarsus as the former contracts.

The palpebral apertures are widened in states of fear or excitement by contraction of the superior and inferior tarsal muscles as a result of increased sympathetic activity. Lesions of the sympathetic supply result in drooping of the upper eyelid (ptosis), as seen in Horner's syndrome.

THE RECTI

The four recti are approximately strap-shaped; each has a thickened middle part that thins gradually to a tendon (Figs 41.8–41.9). They are attached posteriorly to a common tendinous ring that encircles the superior, medial and inferior margins of the optic canal, continues laterally across the inferior and medial parts of the superior orbital fissure, and is attached to a tubercle or spine on the margin of the greater wing of the sphenoid (see Fig. 41.4). The tendinous ring is closely adherent to the dural sheath of the optic nerve medially and to the surrounding periosteum. Inferior rectus, part of medial rectus and the lower fibres of lateral rectus are all attached to the lower part of the ring, whereas superior rectus, part of medial rectus and the upper fibres of lateral rectus are all attached to the upper part. A second small tendinous slip of lateral rectus is attached to the orbital surface of the greater wing of the sphenoid, lateral to the common tendinous ring.

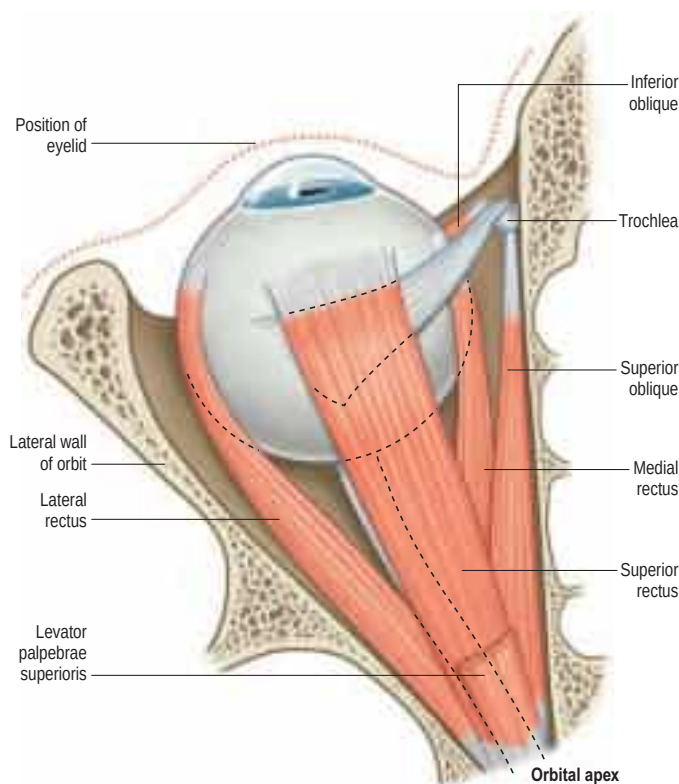


Fig. 41.8 The extraocular muscles viewed from above.

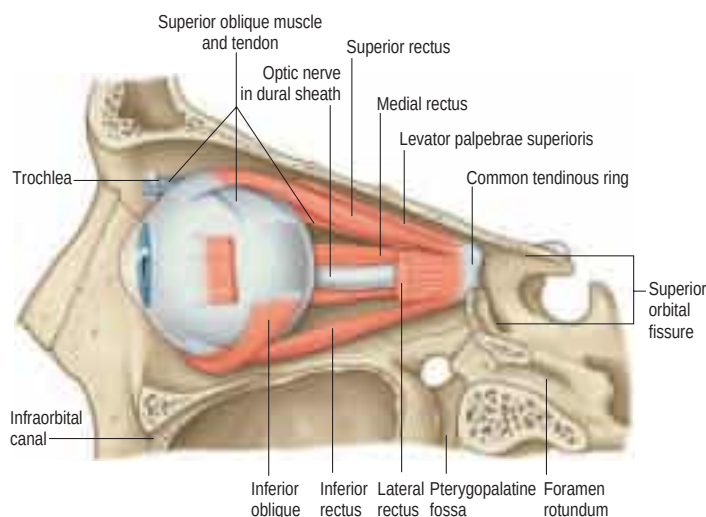


Fig. 41.9 The muscles of the left orbit, lateral view.

Each rectus muscle passes forwards, in the position implied by its name, to be attached anteriorly by a tendinous expansion into the sclera, posterior to the margin of the cornea. However, before their scleral attachment, the recti make functionally important connections within orbital connective tissue that influence muscle action.

Superior rectus

Superior rectus is slightly larger than the other recti. It arises from the upper part of the common tendinous ring, above and lateral to the optic canal. Some fibres also arise from the dural sheath of the optic nerve. The fibres pass forwards and laterally (at an angle of approximately 25° to the median plane of the eye in the primary position) to insert into the upper part of the sclera, approximately 8 mm from the limbus (see Fig. 41.8). The insertion is slightly oblique, the medial margin more anterior than the lateral margin.

Vascular supply Superior rectus receives its arterial supply both directly from the ophthalmic artery and indirectly from its supraorbital branch.

Innervation Superior rectus is innervated by the superior division of the oculomotor nerve that enters the inferior surface of the muscle.

Actions Superior rectus moves the eye so that the cornea is directed upwards (elevation) and medially (adduction). To obtain upward movement alone, the muscle must function with inferior oblique. Superior rectus also causes intorsion of the eye (i.e. medial rotation). Because a check ligament extends from superior rectus to levator palpebrae superioris, elevation of the eye also results in elevation of the upper eyelid. For more detailed discussion of its actions, see page 672.

Inferior rectus

Inferior rectus arises from the common tendinous ring, below the optic canal. It runs along the orbital floor in a similar direction to superior rectus (i.e. forwards and laterally) and inserts obliquely into the sclera below the cornea, approximately 6.5 mm from the limbus (see Fig. 41.9).

Vascular supply Inferior rectus receives its arterial supply from the ophthalmic artery and from the infraorbital branch of the maxillary artery.

Innervation Inferior rectus is innervated by a branch of the inferior division of the oculomotor nerve that enters the superior surface of the muscle.

Actions The principal activity of inferior rectus is to move the eye so that it is directed downwards (depression). It also causes the eye to deviate medially and extorts the eye (i.e. produces lateral rotation). To obtain downward movement alone, inferior rectus must function with superior oblique. A fibrous extension from inferior rectus to the inferior tarsal plate of the eyelid causes the lower eyelid to be depressed when the muscle contracts. For more detailed discussion of its actions, see page 672.

Medial rectus

Medial rectus is slightly shorter than the other recti but is the strongest of the group. It arises from the medial part of the common tendinous ring, and also from the dural sheath of the optic nerve, passing horizontally forwards along the medial wall of the orbit, below superior oblique (see Figs 41.8, 41.9). It inserts into the medial surface of the sclera, approximately 5.5 mm from the limbus and slightly anterior to the other recti.

Vascular supply Medial rectus receives its arterial supply from the ophthalmic artery.

Innervation Medial rectus is innervated by a branch from the inferior division of the oculomotor nerve that enters the lateral surface of the muscle.

Actions Medial rectus moves the eye so that it is directed medially (adduction). The two medial recti acting together are responsible for convergence of the eyes. For more detailed discussion of its actions, see page 672.

Lateral rectus

Lateral rectus arises principally from the lateral part of the common tendinous ring and bridges the superior orbital fissure (see Fig. 41.4); some fibres also arise from a spine on the greater wing of the sphenoid.

The muscle passes horizontally forwards along the lateral wall of the orbit to insert into the lateral surface of the sclera, approximately 7 mm from the limbus (see Fig. 41.8).

Vascular supply Lateral rectus receives its arterial supply from the ophthalmic artery directly and/or from its lacrimal branch.

Innervation Lateral rectus is innervated by the abducens nerve by branches that enter the medial surface of the muscle.

Actions Lateral rectus moves the eye so that it is directed laterally (abducted). For more detailed discussion of its actions, see page 672.

THE OBLIQUES

Superior oblique

Superior oblique is a fusiform muscle that arises from the body of the sphenoid superomedial to the optic canal and the tendinous attachment of the superior rectus (see Fig. 41.4). It runs forwards to end in a round tendon that passes through a fibrocartilaginous loop, the trochlea, attached to the trochlear fossa of the frontal bone (see Fig. 41.8). Tendon and trochlea are separated by a delicate synovial sheath. The tendon subsequently descends posterolaterally and inferior to superior rectus, and is attached to the sclera in the superolateral part of the posterior quadrant behind the equator, between the superior and lateral recti.

Vascular supply Superior oblique receives its arterial supply directly from the ophthalmic artery and indirectly from its supraorbital branch.

Innervation Superior oblique is innervated by the trochlear nerve, which enters the superior surface of the muscle.

Actions Superior oblique is inserted into the posterior part of the eyeball; when it contracts, the back of the eyeball is elevated, and the front of the eyeball is depressed (particularly in the adducted position). Superior oblique also moves the eye laterally (abducted) and intorts the eyeball. For more detailed discussion of its actions, see page 672.

Inferior oblique

Inferior oblique is a thin, narrow muscle that lies near the anterior margin of the floor of the orbit. It arises from the orbital surface of the maxilla lateral to the nasolacrimal fossa and ascends posterolaterally, at first between inferior rectus and the orbital floor, and then between the eyeball and lateral rectus. It is inserted into the lateral part of the sclera behind the equator of the eyeball, in the inferolateral part of the posterior quadrant between the inferior and lateral recti, near, but slightly posterior to, the attachment of superior oblique (see Fig. 41.9). The muscle broadens and thins, and, in contrast to the other extraocular muscles, its tendon is barely discernible at its scleral attachment.

Vascular supply Inferior oblique receives its arterial supply from the ophthalmic artery and from the infraorbital branch of the maxillary artery.

Innervation Inferior oblique is innervated by a branch of the inferior division of the oculomotor nerve that enters the orbital surface of the muscle.

Actions Inferior oblique is inserted into the posterior part of the eyeball; when it contracts, the back of the eyeball is depressed and the front of the eyeball is elevated (particularly in the adducted position). The muscle moves the eye laterally (abduction) and also causes extorsion. For more detailed discussion of its actions, see below.

MINOR MUSCLES OF THE EYELIDS

Several smooth muscles are associated with the orbit, although they are not directly attached to the eyeball. Orbitalis, the orbital muscle of Müller, lies at the back of the orbit and spans the infraorbital fissure. Its functions are uncertain but its contraction may possibly produce a slight forward protrusion of the eyeball. The superior and inferior tarsal muscles are small muscle laminae inserted into the upper and lower eyelids, and are described in more detail with the tarsal plates. All three minor muscles receive a sympathetic innervation from the superior cervical ganglion via the internal carotid plexus.

MOVEMENTS OF THE EYES

Movements of the eyes involve rotations around a centre of rotation within the globe. For practical purposes, this can be considered to lie 13.5 mm behind the corneal apex. Normal eye movements are binocular. Movements of the eyes in the same direction are termed versions, whilst those in opposite directions are termed vergences. Eye movements are often accompanied by corresponding movements of the eyelids, particularly in upgaze, where the activity of levator palpebrae superioris is closely coupled to that of superior rectus. The following section describes the ocular motor system in terms of the actions of individual extraocular muscles, the diversity of eye movements and their neural control.

Actions of the extraocular muscles

Levator palpebrae superioris elevates the upper lid, and its antagonist is the palpebral part of orbicularis oculi. The degree of elevation, which, apart from blinking, is maintained for long periods during waking hours, is a compromise between ensuring an adequate exposure of the cornea and controlling the amount of incident light. In bright sunshine, the latter can be reduced by lowering the upper lid, so limiting glare. The role of the superior tarsal muscle is less clear. Its tonus is related to sympathetic activity, and since ptosis is a consequence of impairment of its sympathetic nerve supply, it may function as an accessory elevator of the upper eyelid.

Six extraocular muscles rotate the eyeball in directions that depend on the geometrical relation between their bony and global attachments (Fig. 41.10), which are altered by the ocular movements themselves. For convenience, each muscle will be considered in isolation, but it must be appreciated that any movement of the eyeball alters the tension and/or length in all six muscles. It is useful to consider the four recti and two obliques as separate groups (remembering always that they act in concert) because they form more obvious groupings as antagonists or synergists. The extrinsic ocular muscles collectively position the eyeball in the orbital cavity and prevent its anteroposterior movements, other than a slight retraction during blinks, because the recti exert a posterior traction while the obliques pull the eyeball to some degree anteriorly. They may be assisted by various 'check ligaments' (see above). A simplified description of the actions of the extraocular muscles is summarized in Figure 41.11.

Of the four recti, the medial and lateral exert comparatively straightforward forces on the eyeball. Being approximately horizontal, when the visual axis is in its primary position, i.e. directed to the horizon, they rotate the eye medially (adduction) or laterally (abduction) about an imaginary vertical axis. They are antagonists. The visual axis can be swept through a horizontal arc by reciprocal adjustment of their lengths. When, as is usual, both eyes are involved, the medial and lateral recti of each eye either can adjust both visual axes in a conjugate movement from point to point at infinity (their axes remaining parallel), or they can converge or diverge the axes to or from nearer or more distant objects of attention in the visual field.

The medial and lateral recti do not rotate the eye around its horizontal axis and so cannot elevate or depress the visual axes as gaze is transferred from nearer to more distant objects or the reverse. This movement requires the superior and inferior recti (aided by the two oblique muscles). It is important to remember that the orbital axis does not correspond with the visual axis in its primary position but diverges from it at an angle of approximately 23° (see Fig. 41.10A) (the value varies between individuals, and depends on the angle between the orbital axes and the median plane). Thus, the simple rotation caused by an isolated superior rectus, analysed with reference to the three hypothetical ocular axes, appears more complex, being primarily elevation (horizontal axis), and secondarily a less powerful medial rotation (vertical axis) and slight intorsion (anteroposterior axis) in which the midpoint of the upper rim of the cornea (often referred to as '12 o'clock') is rotated medially. These actions, compounded as a single, simple rotation, are easily appreciated when it is seen that the direction of traction of superior rectus runs in a posteromedial direction from its attachment in front, which is anterior to the equator and superior to the cornea, to its bony attachment near the orbital apex (see Fig. 41.10). Inferior rectus pulls in a similar direction to superior rectus but rotates the visual axis downwards about the horizontal axis. It rotates the eye medially on a vertical axis but its action around the anteroposterior axis extorts the eye, i.e. rotates it so that the corneal '12 o'clock' point turns laterally. The combined, equal contractions of the superior and inferior recti therefore rotate the eyeball medially, since their effects around the horizontal and anteroposterior axes are opposed. In binocular move-

ments, they assist the medial recti in converging the visual axes, and by reciprocal adjustment can elevate or depress the visual axes. As the eyeball is rotated laterally, the lines of traction of the superior and inferior recti approach the plane of the anteroposterior ocular axis (see Fig. 41.10), and so their rotational effects about this and the vertical ocular axis diminish. In abduction to approximately 23° , they become almost purely an elevator and depressor, respectively, of the visual axis.

Superior oblique acts on the eye from the trochlea and, since the attachment of inferior oblique is, for practical purposes, vertically below this, both muscles approach the eyeball at the same angle, being attached in approximately similar positions in the superior and inferior posterolateral ocular quadrants (see Fig. 41.10). Superior oblique elevates the posterior aspect of the eyeball, and inferior oblique depresses it, which means that the former rotates the visual axis downwards and the latter rotates it upwards, and both movements occur around the horizontal axis. When the eye is in the primary position, the obliquity of both muscles means that they pull in a direction posterior to the vertical axis and both therefore rotate the eye laterally around this axis. With regard to the anteroposterior axis, in isolation, superior oblique intorts the eye and inferior oblique extorts it. Like the superior and inferior recti, therefore, the two obliques have a common turning movement around the vertical axis but are opposed forces in respect of the other two. Acting in concert, they could therefore assist the lateral rectus in abducting the visual axis, as in divergence of the eyes in transferring attention from near to far. Again, like the superior and inferior recti, the directions of traction of the oblique muscles also vary with ocular position, such that they become more nearly a pure elevator and a depressor as the eye is adducted.

Ocular rotations are, for the most part, under voluntary control, whereas torsional movements cannot be initiated at will. When the head is tilted in a frontal plane, reflex torsions occur. Any small lapse in the concerted adjustment of both eyes produces diplopia.

Movements that shift or stabilize gaze

The role of eye movements is to bring the image of objects of visual interest on to the fovea of the retina and to hold the image steady in order to achieve the highest level of visual acuity. Several types of eye movement are required to ensure that these conditions are met. Moreover, the movements of both eyes must be near perfectly matched to achieve the benefits of binocularity. Both voluntary and reflex movements are involved and may be so classified. Alternatively, they may be grouped into those movements that shift gaze as visual interest changes, and those that stabilize gaze by maintaining a steady image on the retina. They have distinct characteristics, and are generated by different neural mechanisms in response to different stimuli, but share a common final motor pathway. Movements that shift or stabilize gaze include saccades, vergence, pursuit and vestibular-generated reflexes (Leigh and Zee 2006).

Saccadic activity is almost omnipresent in human vision. Thus, both visual axes are endlessly and rapidly transferred to new points of interest in any part of the visual field. Binocular gaze is frequently made to travel routes of the most variable complexity in examining objects of interest in the field, and both visual axes must be maintained with sufficient accuracy to avoid diplopia. In so-called 'fixation' of a focus of attention, whether uniocular or binocular, the visual axis is not 'fixed' in a perfectly steady manner but undergoes minute, but measurable, flicking (of a few minutes or even seconds of arc) across the true line of fixation. These microsaccades are rapid and surprisingly complex. When interest changes to another feature of the visual scene, the eyes execute a fast or saccadic movement to take up fixation. If the required rotation is small, the saccade is accurate, whereas small supplementary corrective saccades are needed if the shift is substantial. Saccades may also occur in response to other, i.e. non-visual, exteroceptive stimuli (e.g. auditory, tactile or centrally evoked). They may be volitional or reflex. As an example of the latter, in reading a line of print, the eyes make three or four jerky saccades rather than following the line smoothly; the line is usefully imaged only when the eye is stationary, which means that little of the line is seen by the centre of the fovea. In general, reaction times and movements are measured in microseconds, amplitude varies from seconds of arc to many degrees, with an accuracy of 0.2° or better, and the velocity of a large saccade may reach $500^\circ \text{ sec}^{-1}$. The speed of saccades is assured by an initial, slightly excessive, contraction of the appropriate muscles to overcome orbital viscous resistance. The necessary deceleration when the target is fixated is largely dependent on the elasticity of the extraocular muscles and orbital soft tissues, and not on antagonistic muscular activity.

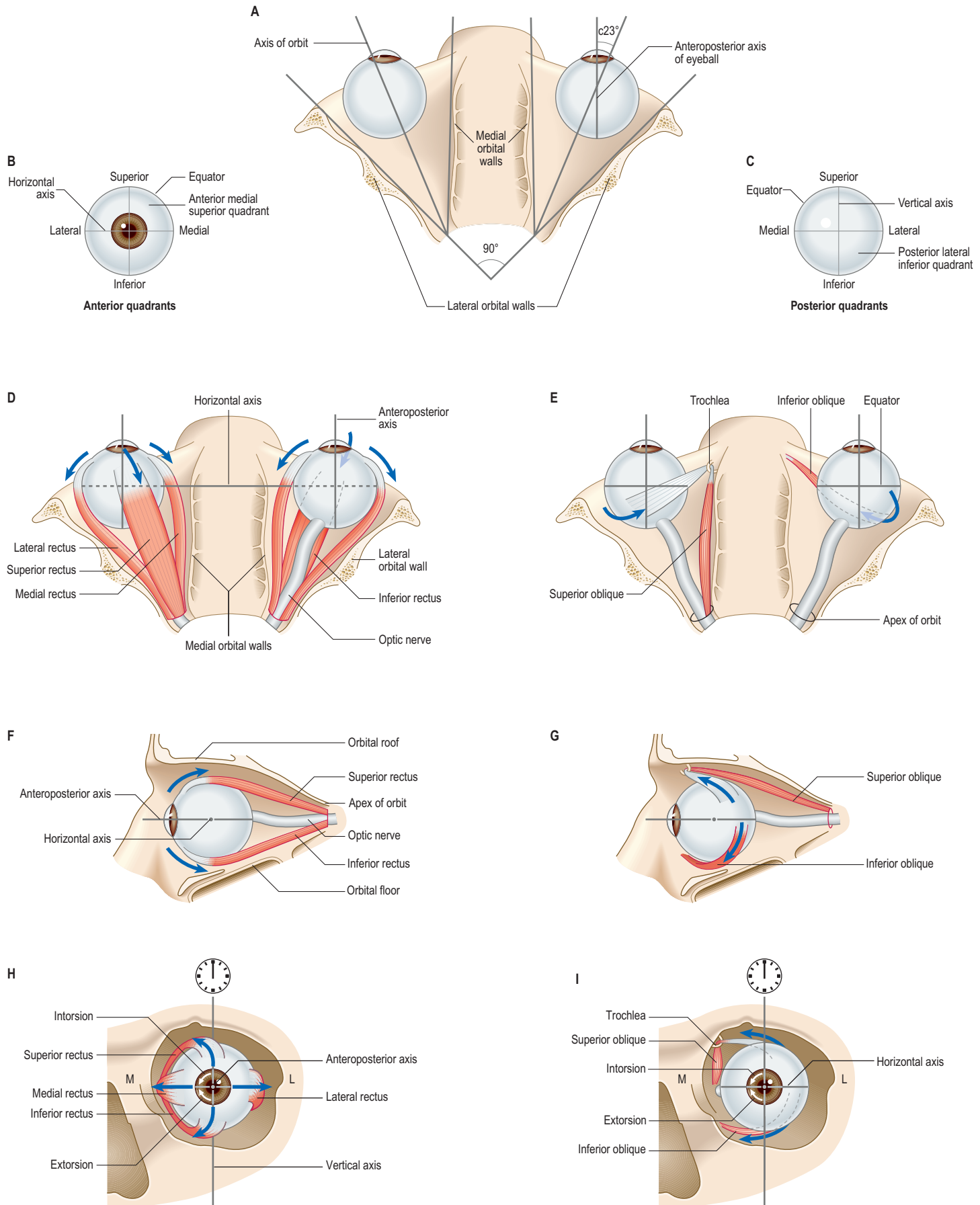


Fig. 41.10 The geometrical basis of ocular movements. **A**, The relationship between the orbital and ocular axes, with the eyes in the primary position, where the visual axes are parallel. **B** and **C**, The ocular globe in anterior and posterior views to show conventional geometry. **D**, The orbits from above, showing the medial and lateral recti and the superior rectus (left) and the inferior rectus (right), indicating turning moments primarily around the vertical axis. **E**, The superior (left) and inferior (right) oblique muscles showing turning moments primarily around the vertical and also anteroposterior axes. **F**, A lateral view to show the actions of the superior and inferior recti around the horizontal axis. **G**, A lateral view to show the action of the superior and inferior oblique muscles around the anteroposterior axis. **H**, An anterior view to show the medial rotational movement of the superior and inferior recti around the vertical axis. Conventionally, the 12 o'clock position indicated is said to be intorted (superior rectus) or extorted (inferior rectus), as indicated by the small arrows on the cornea. **I**, An anterior view to show the torsional effects of the superior oblique (intorsion) and inferior oblique (extorsion) around the anteroposterior axis, as indicated by the small arrows on the cornea.

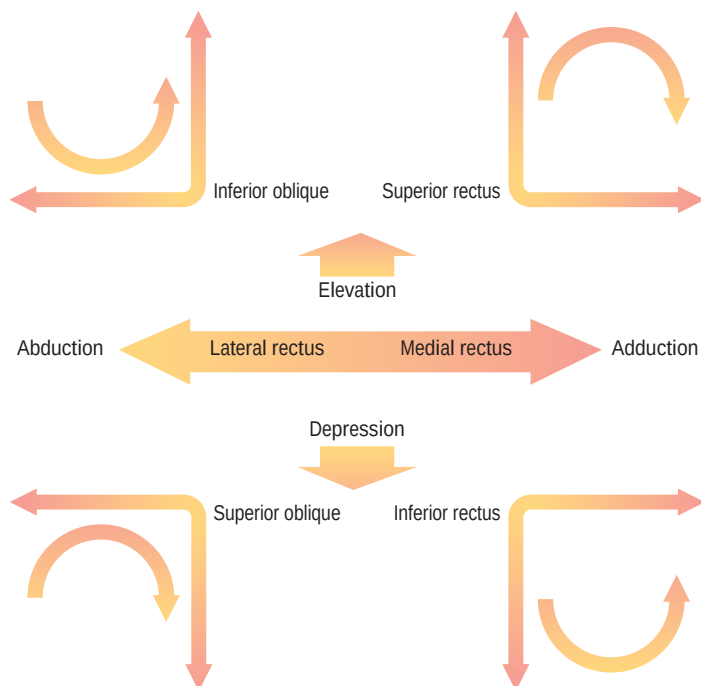


Fig. 41.11 A simplified summary of the actions of the extraocular muscles. Clockwise and anticlockwise curved arrows represent intorsion and extorsion, respectively.

Vergence is a relatively slow movement permitting maintenance of single binocular vision of close objects. The eyes converge towards the midline between the two eyes to achieve imagery of the object on both foveas. The view of the object at the two eyes is not quite the same and the disparity is used to assess depth. In addition, the pupils constrict and the eyes accommodate to achieve sharp, focused images. These three activities constitute the near reflex.

Pursuit eye movements are used to track a moving object of visual interest, maintaining the image approximately on the fovea. They are usually preceded by a saccade to capture the image but, unlike saccades, they are slow and motivated by vision. If the angular shift required to track the moving object is large or the object is moving swiftly, the initial saccade is frequently inaccurate and one or more small corrective saccades are made before tracking begins. Because the stimulus is visual, the pursuit system response is subject to a relatively long latency (approximately 100 msec); the limitation in performance this imposes may be offset by a predictive capacity when object movement follows a regular pattern, and the eye movements adjust in anticipation to speed and direction.

The vestibular apparatus induces a variety of reflex eye movements to compensate for the potentially disruptive effects on vision caused by head and body movement (Ch. 38). Receptors in the semicircular canals respond to active or passive rotational (angular) accelerations of the head. When the body makes substantial rotational movements, a vestibulo-ocular reflex generates a cycle of responses involving both the shifting and stabilizing of gaze. Body rotation is matched by counter-rotation of the eyes so that gaze direction is unaltered and clear vision is maintained. Physical constraint limits the rotation to 30° or less and is followed by a rapid saccadic movement of the eyes to another object in the visual scene and the cycle is repeated. Vision is therefore clear throughout most of the cycle while the image is stationary, but at the cost of no useful vision during the brief periods of the saccades. The reflex is efficient and rapid; this speed could not be generated by the visual system, which is slow relative to the short latency of vestibular receptors.

Other reflexes generated by the vestibular system, which induce compensatory eye movements to stabilize gaze, are activated during brief head movements. When the head is sharply rotated in any direction, the eyeball rotates by an equal amount in the opposite direction in response to the stimulation of semicircular canal cristae (angular acceleration), and gaze is undisturbed. Brief rotational movements are commonly combined with translational movements (linear acceleration) that are monitored by otolith organs. For example, a linear displacement occurs in walking as the head bobs vertically with each stride, and a rotational displacement occurs as the head rolls, invoking otolith and canal responses, respectively, to stabilize the retinal image. Vestibular disease incurring the loss of the rapid, fine compensatory eye

movements in locomotion destabilizes the retinal image, blurs vision and may render locomotion intolerable.

The otoliths also respond to the pull of gravity, generating static vestibulo-ocular reflexes associated with head tilt. When static otolith orientation is changed, e.g. when the head is tilted upwards or downwards, the eyes counter-rotate to maintain fixation of the horizontal meridian. Lateral tilt towards a shoulder generates a torsional counter-rotation of the eyes, a movement that cannot be made voluntarily. The torsional tilt reflex, equal and opposite in direction by the two eyes, is fully compensatory over 40° or so in avoate animals, but in humans it is vestigial; it is fractionally compensatory and varies in extent between individuals. Because the foveal image is unaffected by torsional movements, the subject is unaware of any visual penalty.

The optokinetic response is another visually mediated reflex that stabilizes retinal imagery when a visual scene is rotated about a stationary subject. As the scene changes, the eyes follow and hold the retinal image steady until the eyes shift rapidly in the opposite direction to another area of the visual scene. The full field of vision, rather than small objects within it, is the stimulus, and the alternating slow and fast phases of movement that are generated describe optokinetic nystagmus. This reflex functions in collaboration with the rotational vestibulo-ocular reflex. In sustained rotations of the body, the vestibulo-ocular reflex fades because of the mechanical arrangements of the semicircular canals. In darkness, the reflex, which is initially compensatory, loses velocity, and after approximately 45 seconds the eyeballs become stationary. With a visual input, the reflex is sustained by the optokinetic response. Because the reflex is already initiated, the relative delay of visual input is overcome. The integration of the two systems is served by an accessory visual system projection to the vestibular nuclei via the inferior olive and cerebellum. The usual method of evoking optokinetic nystagmus in the laboratory or clinic is to present a horizontally moving pattern of vertical black-on-white stripes while the head of the subject is held stationary.

Neural control of gaze

Although the detailed anatomical substrates for the different types of eye movement differ, they share common neural circuitry that lies mainly in the pons and midbrain, for horizontal and vertical gaze movements, respectively (Fig. 41.12). The common element for all types of horizontal gaze movements is the abducens nucleus. It contains motor neurones that innervate the ipsilateral lateral rectus and interneurons that project via the medial longitudinal fasciculus to the contralateral oculomotor nucleus, which controls medial rectus. A lesion of the abducens nucleus leads to a total loss of ipsilateral horizontal conjugate gaze. A lesion of the medial longitudinal fasciculus produces slowed or absent adduction of the ipsilateral eye, usually associated with jerky movements (nystagmus) of the abducting eye, a syndrome called internuclear ophthalmoplegia (Leigh and Zee 2006). The gaze motor command involves specialized areas of the reticular formation of the brainstem, which receive a variety of supranuclear inputs. The main region for the generation of horizontal saccades is the paramedian pontine reticular formation, located on each side of the midline in the central paramedian part of the tegmentum, and extending from the pontomedullary junction to the pontopeduncular junction. Each paramedian pontine reticular formation contains excitatory neurones (referred to as 'burst' cells) that discharge at high frequencies just prior to and during ipsilateral saccades. Excitatory burst cells make monosynaptic connections with the ipsilateral abducens nucleus. Pause neurones, located in a midline caudal pontine nucleus, called the nucleus raphe interpositus, discharge tonically during fixation but stop firing immediately prior to a saccade. They appear to exert an inhibitory influence on the burst neurones and act as a switch to change from fixation to saccadic mode (Ramat et al 2007).

The tonic activity of neurones in the nucleus prepositus hypoglossi and medial vestibular nucleus is thought to provide an eye position signal to maintain the eccentric position of the eye against the viscoelastic forces in the orbit. These forces tend to move the eyeball back to looking straight ahead, i.e. the primary position, after a saccade. Vestibular nuclei and the perihypoglossal complex project directly to the abducens nuclei. These projections probably also carry smooth pursuit signals, via the cerebellum.

The final common pathway for vertical gaze movements is formed by the oculomotor and trochlear nuclei. The rostral interstitial nucleus of the medial longitudinal fasciculus contains excitatory burst neurones that discharge in relation to up-and-down vertical saccadic movements and project to motor neurones involved in vertical gaze. The rostral interstitial nucleus of the medial longitudinal fasciculus projects

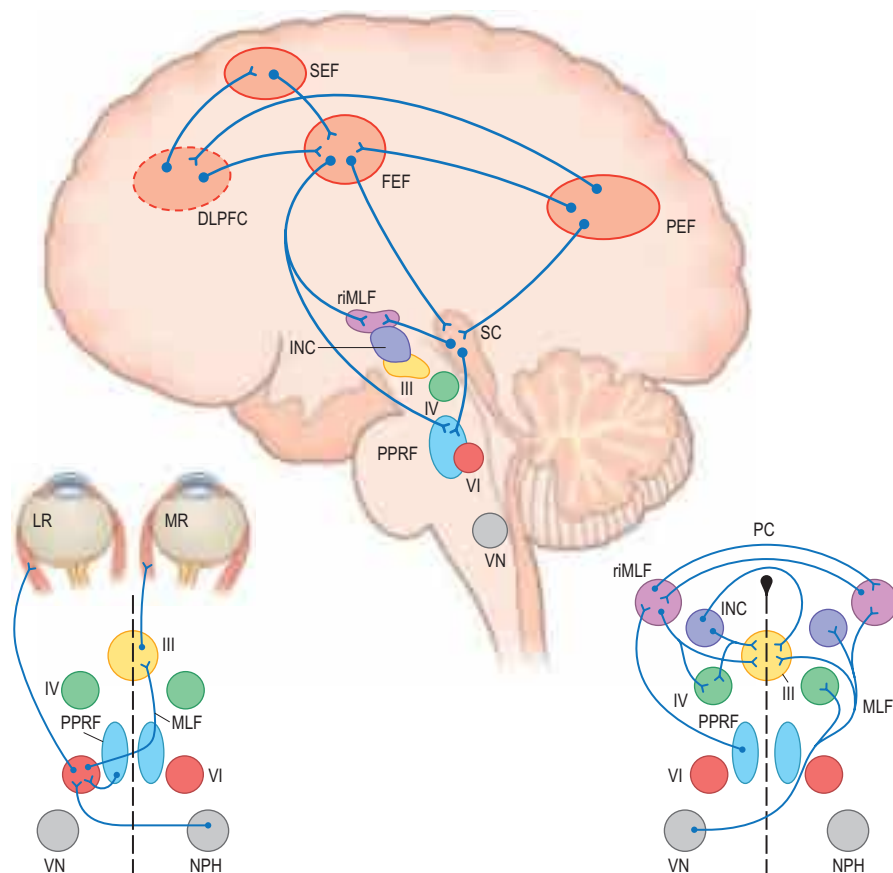


Fig. 41.12 A summary of the control of eye movements. The central drawing shows the supranuclear connections from the frontal eye field (FEF) and the posterior eye field (PEF) to the superior colliculus (SC), rostral interstitial nucleus of the medial longitudinal fasciculus (riMLF), and the paramedian pontine reticular formation (PPRF). The FEF and SC are involved in the production of saccades, while the PEF is thought to be important in the production of pursuit movements. The diagram on the left shows the brainstem pathways for horizontal gaze. Axons from the PPRF travel to the ipsilateral abducens nucleus innervating lateral rectus (LR). Abducens internuclear axons cross the midline and travel in the medial longitudinal fasciculus (MLF) to the neuronal pool in the oculomotor nucleus (III) that innervates medial rectus (MR) of the contralateral eye. The diagram on the right shows the brainstem pathways for vertical gaze. Important structures include the riMLF, PPRF, the interstitial nucleus of Cajal (INC) and the posterior commissure (PC). Other abbreviations: DLPFC, dorsolateral prefrontal cortex; IV, trochlear nucleus; NPH, nucleus propositus hypoglossi; SEF, supplementary eye field; VI, abducens nucleus; VN, vestibular nucleus.

through the posterior commissure to its equivalent on the other side of the mesencephalon, as well as directly to the ipsilateral oculomotor and trochlear nuclei (see Fig. 41.12). Neurones in and around the interstitial nucleus of the medial longitudinal fasciculus, provide signals for vertical gaze holding. Vertical gaze palsies can affect upgaze, downgaze or both. Lesions within the posterior commissure predominantly give rise to disturbances in upgaze, associated with other signs of dorsal mid-brain syndrome, e.g. pupillary abnormalities (light-near dissociation). Discrete lesions placed more ventrally in the region of the rostral interstitial nucleus of the medial longitudinal fasciculus may cause mixed up-and-down, or mainly downgaze disturbances (Leigh and Zee 2006).

The cerebellum plays an important role in the control of eye movements (p. 347, see Fig. 22.24). The vestibulocerebellum (flocculus and nodule) is involved in gaze holding, smooth pursuit and the vestibulo-ocular reflex. The dorsal vermis and fastigial nucleus play a major role in programming accurate saccades and smooth pursuit.

The cerebral hemispheres are extremely important for the programming and coordination of both saccadic and pursuit conjugate eye movements (Ch. 25). There appear to be four main cortical areas in the cerebral hemispheres involved in the generation of saccades. These are the frontal eye field, which is located at the intersection of the superior precentral sulcus with the superior frontal sulcus (Brodmann area 8) (Amiez and Petrides 2009); the supplementary eye field, which lies in the dorsomedial frontal cortex (Brodmann area 6); the dorsolateral prefrontal cortex, which lies anterior to the frontal eye field in the second frontal gyrus (Brodmann area 46); and a posterior eye field, which lies in the parietal lobe in the medial wall of the posterior half of the intraparietal sulcus, including parts of the supramarginal and angular gyri (Brodmann areas 39 and 40), and the adjacent lateral intraparietal sulcus (Müri et al 1996). These areas all appear to be interconnected and to send projections to the superior colliculus and the brainstem areas controlling saccades.

Two parallel pathways are involved in the cortical generation of saccades. An anterior system originates in the frontal eye field and projects, both directly and via the superior colliculus, to the brainstem saccadic generators. This pathway also passes indirectly via the basal ganglia to the superior colliculus. Projections from the frontal cortex influence cells in the pars reticularis of the substantia nigra, via a relay in the caudate nucleus. An inhibitory pathway from the pars reticulata projects directly to the superior colliculus. This may be a gating circuit related

to voluntary saccades, especially of the memory-guided type. A posterior pathway originates in the posterior eye field and passes to the brainstem saccadic generators via the superior colliculus. This pathway is important for triggering visually guided saccades.

The smooth pursuit system has developed relatively independently of the saccadic oculomotor system to maintain foveation of a moving target, although there are inevitable interconnections between the two. The first task is to identify and code the velocity and direction of a moving target. This is carried out in the extrastriate visual area known as the middle temporal visual area (also called visual area V5), which contains neurones sensitive to visual target motion. In humans, this lies immediately posterior to the ascending limb of the inferior temporal sulcus at the occipitotemporal border. The middle temporal visual area sends this motion signal to the medial superior temporal visual area, thought to lie superior and a little anterior to the middle temporal visual area within the inferior parietal lobe; damage to this area results in an impairment of smooth pursuit of targets moving towards the damaged hemisphere.

Both the medial superior temporal visual area and the frontal eye field send direct projections to a group of nuclei that lie in the basal part of the pons. In monkeys, the dorsolateral and lateral groups of pontine nuclei receive direct cortical inputs related to smooth pursuit. Lesions of similarly located nuclei in humans result in abnormal pursuit. These nuclei transfer the pursuit signal bilaterally to the posterior vermis, contralateral flocculus and fastigial nuclei of the cerebellum. The pursuit signal ultimately passes from the cerebellum to the brainstem, specifically to the medial vestibular nucleus and nucleus propositus hypoglossi, and thence to the paramedian pontine reticular formation and possibly directly to the ocular motor nuclei. This circuitry therefore involves a double decussation: firstly, at the level of the midpons (pontocerebellar neurones), and secondly, in the lower pons (vestibulo-abducens neurones).

The vestibulo-ocular reflex maintains coordination of vision during movement of the head and results in a compensatory conjugate eye movement that is equal but opposite to the movement of the head. This essentially three-neurone arc consists of primary vestibular neurones that project to the vestibular nuclei, secondary neurones that project from these nuclei directly to the abducens and oculomotor nuclei, and tertiary neurones that innervate the extraocular muscles. Vestibular neurones responding to head rotation also respond to optokinetic stimuli, which means that the neural substrate is likely to include both the visual and vestibular systems.

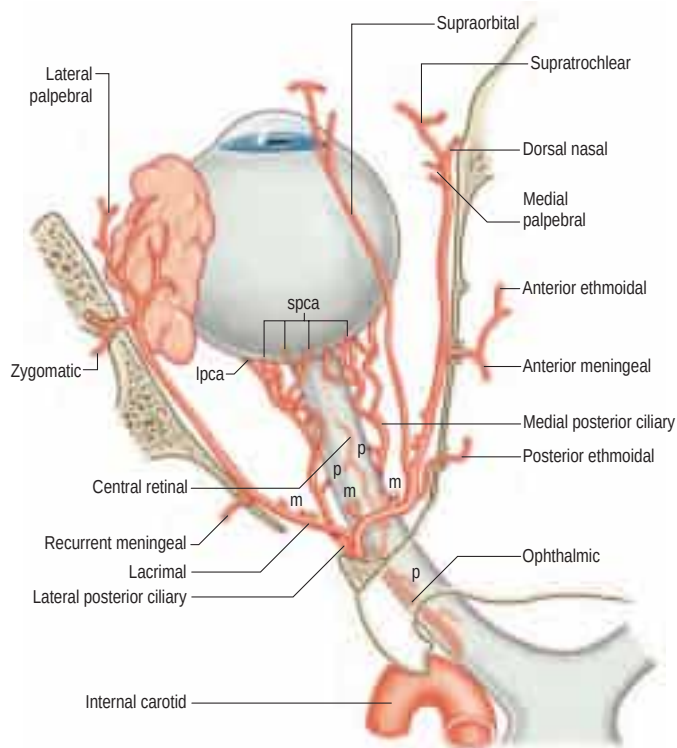


Fig. 41.13 The orbital distribution of the ophthalmic artery. Abbreviations: lpca, long posterior ciliary artery; m, muscular arteries; p, pial arteries; spca, short posterior ciliary arteries.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

The main vessel supplying orbital structures is the ophthalmic artery (Fig. 41.13). Its terminal branches anastomose on the face and scalp with those of the facial, maxillary and superficial temporal arteries, thereby establishing connections between the external and internal carotid arteries. The infraorbital branch of the maxillary artery, and possibly the recurrent meningeal artery, also supply orbital structures.

Ophthalmic artery

The ophthalmic artery leaves the internal carotid artery as it exits the cavernous sinus medial to the anterior clinoid process. It enters the orbit by the optic canal, inferolateral to the optic nerve and continues forwards for a short distance before turning medially by crossing (almost always) over or (sometimes) under the optic nerve (see Fig. 41.13). The main trunk of the artery continues along the medial wall of the orbit between the superior oblique and lateral rectus, and divides into supraorbital (frontal) and dorsal nasal branches at the medial end of the upper eyelid. Although the order of branches from the ophthalmic artery is quite variable, a number may be identified consistently, including the central retinal artery, lacrimal artery, muscular branches, ciliary arteries, supraorbital artery, anterior and posterior ethmoidal arteries, meningeal branch, medial palpebral arteries, supratrochlear artery and dorsal nasal artery (Hayreh and Dass 1962). Many of the branches of the ophthalmic artery accompany sensory nerves of the same name and have a similar distribution. Variations in the origin and course of the ophthalmic artery have been described (Huynh-Le et al 2005). The artery may arise within, near or significantly distal to the upper dural ring, and the point of penetration of the dura lining the optic canal may be posterior, below or anterior to the falciform ligament. The ophthalmic artery may also arise from the clinoid or intracavernous segment of the carotid artery. In the latter case, the ophthalmic artery may pass through either the superior orbital fissure or the optic canal.

Central retinal artery

The small central artery of the retina is the first branch of the ophthalmic artery. It begins below the optic nerve and, for a short distance, lies in the dural sheath of the nerve. It enters the inferomedial surface of the nerve 6.4–15.2 mm behind the eye, and runs to the retina (accompanying the central retinal vein) along its axis. The further distribution of the central retinal artery is described on page 704.

Muscular branches

Muscular branches frequently spring from a common trunk to form superior and inferior groups, most of which accompany branches of the oculomotor nerve. The inferior branch contains most of the anterior ciliary arteries. Other muscular vessels branch from the lacrimal and supraorbital arteries or from the trunk of the ophthalmic artery.

Ciliary arteries

The ciliary arteries are distributed in long and short posterior, and in anterior groups (see Fig. 41.13). Long posterior ciliary arteries, usually two, pierce the sclera near the optic nerve, pass anteriorly along the horizontal meridian and join the major arterial circle of the iris (see Fig. 42.8). About seven short posterior ciliary arteries pass close to the optic nerve to reach the eyeball, where they divide into 15–20 branches. They pierce the sclera around the optic nerve to supply the choroid, and anastomose with twigs of the central retinal artery at the optic disc (see Fig. 42.30). Anterior ciliary arteries arise from muscular branches of the ophthalmic artery. They reach the eyeball on the tendons of the recti, form a circumcorneal subconjunctival vascular zone, and pierce the sclera near the sclerocorneal junction to end in the major arterial circle of the iris.

Lacrimal artery

The lacrimal artery is a large branch that usually leaves the ophthalmic artery near its exit from the optic canal (see Fig. 41.13), although it occasionally arises before the ophthalmic artery enters the orbit. It accompanies the lacrimal nerve along the upper border of lateral rectus, supplies and traverses the lacrimal gland, and ends in the eyelids and conjunctiva as the lateral palpebral arteries. The latter run medially in the upper and lower lids and anastomose with the medial palpebral arteries. The lacrimal artery gives off one or two zygomatic branches. One reaches the temporal fossa via the zygomaticotemporal foramen and anastomoses with the deep temporal arteries. The other reaches the cheek by the zygomaticofacial foramen and anastomoses with transverse facial and zygomatico-orbital arteries. A recurrent meningeal branch, usually small, passes back via the lateral part of the superior orbital fissure to anastomose with a middle meningeal branch. This branch is sometimes large and replaces the lacrimal artery, in which case it becomes a more significant contributor to the orbital blood supply.

Supraorbital artery

The supraorbital artery leaves the ophthalmic artery where it crosses the optic nerve (see Fig. 41.13), and ascends medial to superior rectus and levator palpebrae superioris. It accompanies the supraorbital nerve between the periosteum and levator palpebrae superioris, passes through the supraorbital foramen or notch, and divides into superficial and deep branches. These supply the skin, muscles and frontal periosteum, and anastomose with the supratrochlear artery and with the frontal branch of the superficial temporal artery and its contralateral fellow (see Fig. 30.20). The supraorbital artery supplies superior rectus and levator palpebrae superioris, and sends a branch across the trochlea to the medial canthus. It often sends a branch to the diploë of the frontal bone at the supraorbital margin, and may also supply the mucoperiosteum in the frontal sinus.

Posterior ethmoidal artery

The posterior ethmoidal artery runs through the posterior ethmoidal canal and supplies the posterior ethmoidal air sinuses. Entering the cranium, it gives off a meningeal branch to the dura mater, and nasal branches that descend into the nasal cavity via the cribriform plate and anastomose with branches of the sphenopalatine artery (see Fig. 33.9).

Anterior ethmoidal artery

The anterior ethmoidal artery passes with its accompanying nerve through the anterior ethmoidal canal to supply the ethmoidal and frontal air sinuses. Entering the cranium, it gives off a meningeal branch to the dura mater, and nasal branches that descend into the nasal cavity with the anterior ethmoidal nerve (see Fig. 33.9). It runs in a groove on the deep surface of the nasal bone to supply the lateral nasal wall and septum. A terminal branch appears on the nose between the nasal bone and the upper nasal cartilage (see Fig. 33.9).

Meningeal branch

A meningeal branch, usually small, passes back through the superior orbital fissure to the middle cranial fossa, where it anastomoses with the middle and accessory meningeal arteries. It is sometimes large, in which case it becomes a major contributor to the orbital blood supply.

Medial palpebral arteries

Above the trochlea, two medial palpebral arteries branch separately from the ophthalmic artery and join the lateral palpebral arteries to complete the arcades of the upper and lower eyelids.

Supratrochlear artery

The supratrochlear artery is a terminal branch of the ophthalmic artery. It leaves the orbit superomedially with the supratrochlear nerve, ascends on the forehead to supply the skin, muscles and pericranium, and anastomoses with the supraorbital artery and with its contralateral fellow.

Dorsal nasal artery

The dorsal nasal artery is the other terminal branch of the ophthalmic artery and emerges from the orbit between the trochlea and medial palpebral ligament. It gives a branch to the upper part of the nasolacrimal sac and then divides into two branches. One branch joins the terminal part of the facial artery, and the other runs along the dorsum of the nose, supplies its outer surface and anastomoses with its contralateral fellow and the lateral nasal branch of the facial artery.

Infraorbital branch of the maxillary artery

The infraorbital branch of the maxillary artery enters the orbit through the inferior orbital fissure (see Fig. 32.20). Accompanying the infraorbital nerve, it passes along the infraorbital groove of the maxilla in the floor of the orbit before entering the infraorbital canal, and comes out on to the face through the infraorbital foramen. While in the infraorbital groove, it gives off branches that supply inferior rectus and inferior oblique, the nasolacrimal sac and, occasionally, the lacrimal gland.

VEINS

The orbit is drained by the superior and inferior ophthalmic veins and the infraorbital vein (Fig. 41.14). The veins of the eyeball mainly drain into the vortex veins; the retinal veins drain into the central retinal vein.

Superior and inferior ophthalmic veins

The superior and inferior ophthalmic veins link the facial and intracranial veins, and are devoid of valves. The superior ophthalmic vein forms posteromedial to the upper eyelid from two tributaries that connect anteriorly with the facial and supraorbital veins. It runs with the ophthalmic artery, lying between the optic nerve and superior rectus, and receives the corresponding tributaries, the two superior vortex veins of the eyeball, and the central vein of the retina. The central retinal vein sometimes drains directly into the cavernous sinus, although it still gives a communicating branch to the superior ophthalmic vein. The

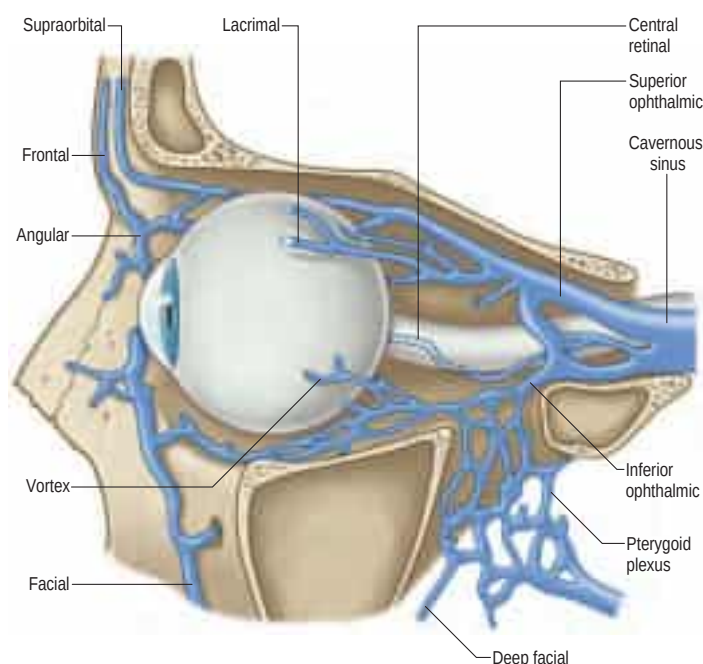


Fig. 41.14 The principal veins of the orbit.

superior ophthalmic vein may also receive the inferior ophthalmic vein. It traverses the superior orbital fissure, usually above the common tendinous ring of the recti, and ends in the cavernous sinus.

The inferior ophthalmic vein begins in a network near the anterior region of the orbital floor and medial wall. It runs backwards on inferior rectus and across the inferior orbital fissure, and then either joins the superior ophthalmic vein or passes through the superior orbital fissure, within or below the common tendinous ring, to drain directly into the cavernous sinus. The inferior ophthalmic vein receives tributaries from inferior rectus and inferior oblique, the nasolacrimal sac, the eyelids and the two inferior vortex veins of the eyeball. It communicates with the pterygoid venous plexus by a branch that passes through the inferior orbital fissure, and may also communicate with the facial vein across the inferior margin of the orbit.

Infraorbital vein

The infraorbital vein runs with the infraorbital nerve and artery in the floor of the orbit, and passes backwards through the inferior orbital fissure into the pterygoid venous plexus. It drains structures in the floor of the orbit and communicates with the inferior ophthalmic vein; it may also communicate with the facial vein on the face.

LYMPHATIC DRAINAGE

Lymphatic vessels other than those draining the conjunctiva have not been identified.

INNERVATION

Somatic and autonomic motor and somatic sensory nerves are found in the orbit (Figs 41.15–41.17). The oculomotor, trochlear and abducens nerves supply the extraocular muscles. Parasympathetic fibres from the oculomotor nerve supply sphincter pupillae and the ciliary muscle (ciliaris) via the ciliary ganglion, and those from the facial nerve innervate the lacrimal gland and choroid via the pterygopalatine ganglion. Sympathetic fibres supply dilator pupillae. Both sympathetic and parasympathetic nerves supply the arteries. The sensory nerves within the orbit are the optic, ophthalmic and maxillary nerves (the maxillary nerve and most of the ophthalmic branches only pass through the orbit *en route* to supply the face and jaws).

Oculomotor nerve

The oculomotor nerve is the third cranial nerve (see Fig. 41.15). It innervates levator palpebrae superioris and four of the extraocular muscles (superior, inferior and medial rectus and inferior oblique), and also conveys parasympathetic fibres that relay in the ciliary ganglion. The nerve emerges at the midbrain, on the medial side of the crus of the cerebral peduncle, and passes along the lateral dural wall of the cavernous sinus, dividing into superior and inferior divisions that run beneath the trochlear and ophthalmic nerves. The two divisions enter the orbit through the superior orbital fissure, within the common

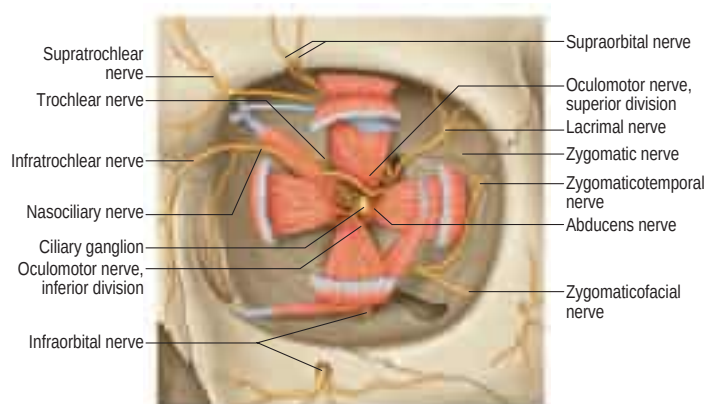


Fig. 41.15 The left orbit viewed from in front, showing motor and sensory nerves and the origins of the extraocular muscles. Note that in this figure the trochlear nerve is shown inserting on the inferior aspect of the superior oblique, whereas it more commonly inserts on the superior surface of the muscle. (With permission from Dutton JJ 1994 Atlas of Clinical and Surgical Orbital Anatomy. Philadelphia: Saunders.)

tendinous ring of the recti, separated by the nasociliary branch of the ophthalmic nerve.

The superior division of the oculomotor nerve passes above the optic nerve to enter the inferior (ocular) surface of superior rectus. It supplies this muscle and gives off a branch that runs to innervate levator palpebrae superioris. The inferior division of the oculomotor nerve divides into medial, central and lateral branches. The medial branch passes beneath the optic nerve to enter the lateral (ocular) surface of medial rectus; the central branch runs downwards and forwards to enter the superior (ocular) surface of inferior rectus; the lateral branch travels forwards on the lateral side of inferior rectus to enter the orbital surface of inferior oblique and also communicates with the ciliary ganglion to distribute parasympathetic fibres to sphincter pupillae and the ciliary muscle.

Trochlear nerve

The trochlear nerve is the fourth cranial nerve and innervates superior oblique exclusively. It is the only cranial nerve to emerge from the dorsal surface of the brainstem, passing from the midbrain on to the lateral surface of the crus of the cerebral peduncle. It runs through the lateral dural wall of the cavernous sinus and then crosses the oculomotor nerve to enter the orbit through the superior orbital fissure, outside the common tendinous ring, above levator palpebrae superioris and medial to the frontal and lacrimal nerves (see Fig. 41.16). The trochlear nerve travels a short distance to enter the superior (orbital) surface of superior oblique.

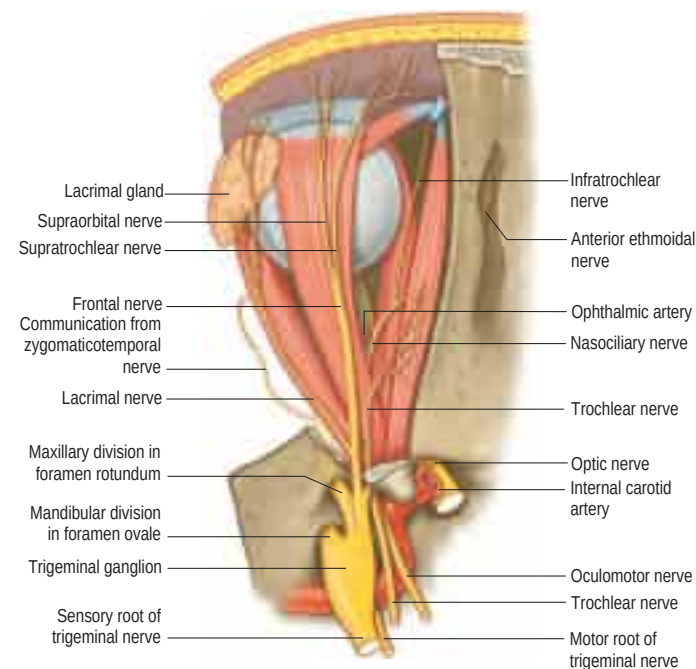


Fig. 41.16 The nerves of the left orbit, superior aspect.

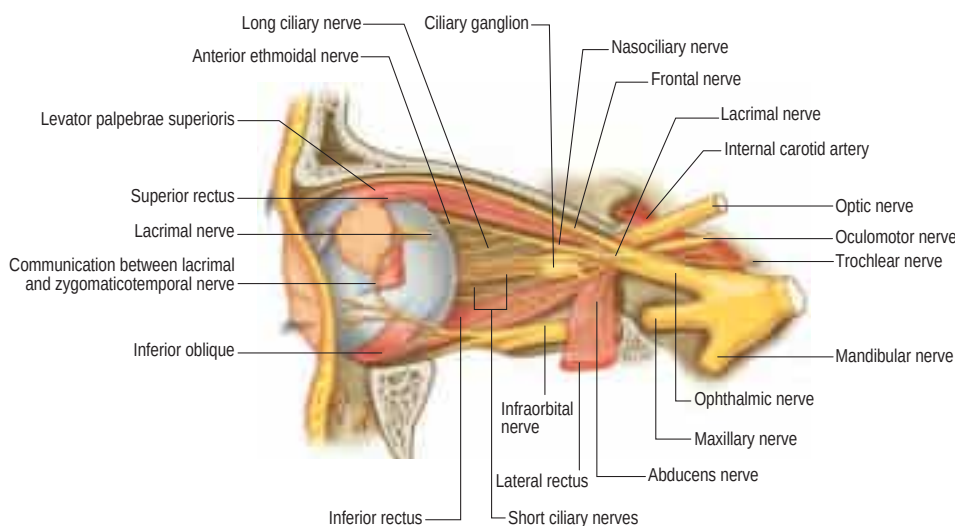


Fig. 41.17 The nerves of the left orbit and the ciliary ganglion, lateral aspect.

Abducens nerve

The abducens nerve is the sixth cranial nerve and exclusively innervates lateral rectus. It emerges from the brainstem between the pons and the medulla oblongata, and usually runs through the inferior venous compartment of the petroclival venous confluence in a bow-shaped canal, Dorello's canal. It then bends sharply across the upper border of the petrous part of the temporal bone to enter the cavernous sinus, where it lies lateral to the internal carotid artery (unlike the oculomotor, trochlear, ophthalmic and maxillary nerves, which merely invaginate the lateral dural wall of the sinus). The abducens nerve enters the orbit through the superior orbital fissure, within the common tendinous ring, at first below, and then between, the two divisions of the oculomotor nerve and lateral to the nasociliary nerve (see Fig. 41.15). It passes forwards to enter the medial (ocular) surface of lateral rectus.

Optic nerve

The optic nerve is the second cranial nerve. It arises from the optic chiasma on the floor of the diencephalon and enters the orbit through the optic canal, accompanied by the ophthalmic artery. It changes its shape from being flattened at the chiasma to rounded as it passes through the optic canal. In the orbit, the optic nerve passes forwards, laterally and downwards, and pierces the sclera at the lamina cribrosa, slightly medial to the posterior pole. It has a somewhat tortuous course within the orbit to allow for movements of the eyeball, and is surrounded by extensions of the three layers of the meninges.

The optic nerve has important relationships with other orbital structures (see Fig. 41.4). As it leaves the optic canal, it lies superomedial to the ophthalmic artery, and is separated from lateral rectus by the oculomotor, nasociliary and abducens nerves, and sometimes by the ophthalmic veins. It is closely related to the origins of the four recti, whereas more anteriorly, where the muscles diverge, it is separated from them by a substantial amount of orbital fat. Just beyond the optic canal, the ophthalmic artery and the nasociliary nerve cross the optic nerve to reach the medial wall of the orbit. The central artery of the retina enters the substance of the optic nerve about halfway along its length. Near the back of the eyeball, the optic nerve becomes surrounded by the long and short ciliary nerves and vessels.

Ophthalmic nerve

The ophthalmic division of the trigeminal nerve arises from the trigeminal ganglion in the middle cranial fossa. It passes forwards along the lateral dural wall of the cavernous sinus, giving off three main branches – the lacrimal, frontal and nasociliary nerves – just before it reaches the superior orbital fissure (see Figs 41.16–41.17). These branches subsequently travel through the orbit to supply targets that are primarily in the upper part of the face (see Fig. 30.22).

Lacrimal nerve

The lacrimal nerve enters the orbit through the superior orbital fissure, outside the common tendinous ring and lateral to the frontal and trochlear nerves. It passes forwards along the lateral wall of the orbit on the superior border of lateral rectus, and travels through the lacrimal

gland and the orbital septum to supply the conjunctiva and skin covering the lateral part of the upper eyelid. The nerve also communicates with the zygomatic branch of the maxillary nerve, which may carry some postganglionic parasympathetic fibres from the pterygopalatine ganglion to the lacrimal gland (see orbital branches from pterygopalatine ganglion, below).

Frontal nerve

The frontal nerve is the largest branch of the ophthalmic nerve. It enters the orbit through the superior orbital fissure outside the common tendinous ring, and lies between the lacrimal nerve laterally and the trochlear nerve medially. It passes forwards towards the rim of the orbit on levator palpebrae superioris; about halfway along this course, it divides into the supraorbital and supratrochlear nerves.

Supraorbital nerve

The supraorbital nerve, the larger of the two terminal branches of the frontal nerve, continues forwards along levator palpebrae superioris until it leaves the orbit through the supraorbital notch or foramen. It emerges on to the forehead and supplies the mucous membrane that lines the frontal sinus, the conjunctiva and skin covering the upper eyelid, and the skin over the forehead and scalp. The postganglionic sympathetic fibres that innervate the sweat glands of the supraorbital area probably travel in the supraorbital nerve, having entered the ophthalmic nerve via its communication with the abducens nerve within the cavernous sinus.

Supratrochlear nerve

The supratrochlear nerve runs medially above the trochlea for the tendon of superior oblique. It gives a descending branch to the infratrochlear nerve and ascends on to the forehead through the frontal notch to supply the conjunctiva and skin covering the upper eyelid and the forehead.

Nasociliary nerve

The nasociliary nerve is intermediate in size between the frontal and lacrimal nerves, and is more deeply placed in the orbit, which it enters through the common tendinous ring, lying between the two rami of the oculomotor nerve. It crosses the optic nerve with the ophthalmic artery and runs obliquely below superior rectus and superior oblique to reach the medial orbital wall, where it gives off the anterior and posterior ethmoidal nerves. The nasociliary nerve also has long ciliary and infratrochlear branches and a connection with the ciliary ganglion.

Anterior ethmoidal nerve

The anterior ethmoidal nerve passes through the anterior ethmoidal foramen and canal, and enters the cranial cavity. It runs forwards in a groove on the upper surface of the cribriform plate beneath the dura mater, and descends through a slit lateral to the crista galli into the nasal cavity, where it occupies a groove on the internal surface of the nasal bone and gives off the medial and lateral internal nasal branches. For a description of the subsequent distribution of the anterior ethmoidal nerve, see [page 564](#).

Posterior ethmoidal nerve

The posterior ethmoidal nerve leaves the orbit by the posterior ethmoidal foramen and supplies the ethmoidal and sphenoidal sinuses.

Infratrochlear nerve

The infratrochlear nerve leaves the orbit below the trochlea and supplies the skin of the eyelids, the conjunctiva, lacrimal sac, lacrimal caruncle and the side of the nose above the medial canthus.

Long ciliary nerves

Two or three long ciliary nerves branch from the nasociliary nerve as it crosses the optic nerve. They accompany the short ciliary nerves, pierce the sclera near the attachment of the optic nerve, and run forwards between the sclera and choroid. They supply the ciliary body, iris and cornea, and contain postganglionic sympathetic fibres for the dilator pupillae from neurones in the superior cervical ganglion. An alternative pathway for the supply of the dilator pupillae is via the sympathetic root associated with the ciliary ganglion ([Fig. 41.18](#)).

Ramus communicans to the ciliary ganglion

The ramus communicans to the ciliary ganglion usually branches from the nasociliary nerve as the latter enters the orbit lateral to the optic

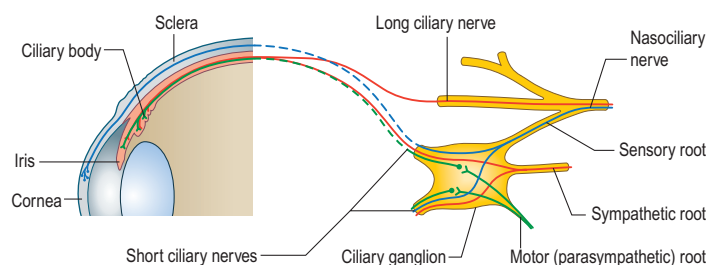


Fig. 41.18 The ciliary ganglion, with its roots and branches of distribution. Key: red, sympathetic fibres; green, parasympathetic fibres; blue, sensory fibres. Alternative pathways are given for the sympathetic fibres.

nerve. It is sometimes joined by a filament from the internal carotid sympathetic plexus or from the superior ramus of the oculomotor nerve as it enters the posterosuperior angle of the ganglion.

Maxillary nerve

Most of the branches of the maxillary division of the trigeminal nerve arise in the pterygopalatine fossa. They include the zygomatic and infraorbital nerves, which pass into the orbit through the inferior orbital fissure (see [Figs 32.30, 41.17](#)).

Zygomatic nerve

The zygomatic nerve lies close to the base of the lateral wall of the orbit. It divides soon after entering the orbit into two branches, the zygomaticotemporal and the zygomaticofacial nerves, which run within the orbit for only a short distance before passing on to the face through the lateral wall of the orbit. Either they may enter separate canals within the zygomatic bone or the zygomatic nerve itself may enter the bone before dividing. For a description of the distribution of the zygomaticotemporal and zygomaticofacial nerves on the face, see [page 502](#).

Infraorbital nerve

The infraorbital nerve initially lies in the infraorbital groove on the floor of the orbit (see [Fig. 32.30B](#)). As it approaches the rim of the orbit, it runs into the infraorbital canal, through which it passes to emerge on to the face at the infraorbital foramen. It supplies the skin of the lower eyelid and possibly the conjunctiva. For a description of the subsequent distribution of the infraorbital nerve on the face, see [page 502](#).

Ciliary ganglion

The ciliary ganglion is a parasympathetic ganglion concerned with the innervation of certain intraocular muscles. It is a small, flat, reddish-grey swelling, 1–2 mm in diameter, connected to the nasociliary nerve, and located near the apex of the orbit in loose fat approximately 1 cm in front of the medial end of the superior orbital fissure. It lies between the optic nerve and lateral rectus, usually lateral to the ophthalmic artery. Its neurones, which are multipolar, are larger than those found in typical autonomic ganglia; a very small number of more typical neurones are also present.

Its connections or roots (motor, sensory and sympathetic) enter or leave the ganglion posteriorly (see [Fig. 41.18](#)). Eight to ten delicate filaments, termed the short ciliary nerves, emerge anteriorly from the ganglion, arranged in two or three bundles, the lower being larger. They run forwards sinuously with the ciliary arteries, above and below the optic nerve, and divide into 15–20 branches that pierce the sclera around the optic nerve and run in small grooves on the internal scleral surface. They convey parasympathetic, sympathetic and sensory fibres between the eyeball and the ciliary ganglion; only the parasympathetic fibres synapse in the ganglion.

The parasympathetic root, derived from the branch of the oculomotor nerve to the inferior oblique, consists of preganglionic fibres from the Edinger–Westphal nucleus, which relay in the ganglion. Postganglionic fibres travel in the short ciliary nerves to the sphincter pupillae and ciliary muscle. More than 95% of these fibres supply the ciliary muscle, which is much the larger muscle in volume.

The sympathetic root contains fibres from the plexus around the internal carotid artery within the cavernous sinus. These postganglionic fibres, derived from the superior cervical ganglion, form a fine branch that enters the orbit through the superior orbital fissure inside the common tendinous ring. The fibres may pass directly to the ganglion, or may join the nasociliary nerve and travel to the ganglion in its sensory root; either way, they traverse the ganglion without synapsing

to emerge into the short ciliary nerves. They are distributed to the blood vessels of the eyeball. Sympathetic fibres innervating dilator pupillae may sometimes travel via the short ciliary nerves (rather than the more usual route via the ophthalmic, nasociliary and long ciliary nerves).

The sensory fibres that pass through the ciliary ganglion are derived from the nasociliary nerve. They enter the short ciliary nerves and carry sensation from the cornea, the ciliary body and the iris.

Orbital branches of the pterygopalatine ganglion

Several rami orbitales arise dorsally from the pterygopalatine ganglion and enter the orbit through the inferior orbital fissure. Branches leave the orbit through the posterior ethmoidal air sinus. There is strong experimental evidence from studies of animals, including monkeys, that postganglionic parasympathetic branches pass directly to the lacrimal gland, ophthalmic artery and choroid.

EYELIDS, CONJUNCTIVA AND LACRIMAL SYSTEM

The exposed ocular surface is protected by retractable eyelids and by a tear film produced mainly by the lacrimal gland, with contributions from glands within the eyelid and conjunctiva (a transparent mucous membrane that covers the inner surface of the eyelid and the exposed surface of the sclera) (see Fig. 42.1).

EYELIDS

The eyelids (palpebrae) are two folds of modified skin that cover the anterior surface of the eye (Fig. 41.19). By their reflex closure, achieved by contraction of orbicularis oculi, they protect the eye from injury and shield the eyes from excessive light. Periodic blinking maintains a thin film of tears over the cornea that prevents desiccation; movement of the eyelids during blinking helps ensure the even distribution of the tear film and facilitates tear outflow through the nasolacrimal drainage system.

The upper eyelid is larger and more mobile than the lower eyelid and contains an elevator muscle, levator palpebrae superioris (see above). A transverse opening, the palpebral fissure, lies between the free margins of the lids, which join at their extremities (termed the medial and lateral canthus). The lateral canthus is relatively featureless. The medial canthus is approximately 2 mm lower than the lateral canthus; this distance is increased in some Asiatic groups. It is separated from the eyeball by a small triangular space, the lacrimal lake (lacus lacrimalis), in which a small, reddish body called the lacrimal caruncle is situated. The caruncle represents an area of modified skin containing some fine hairs and is mounted on the plica semilunaris, a fold of conjunctiva that is believed by some to be a vestige of the nictitating membrane of other animals.

A small elevation, the lacrimal papilla, is located on each palpebral margin approximately one-sixth of the way along from the medial canthus of the eye. There is a small aperture, the punctum lacrimale, in the centre of the papilla that forms the opening to the lacrimal drainage system. The margin of the eyelid lateral to the lacrimal papilla bears

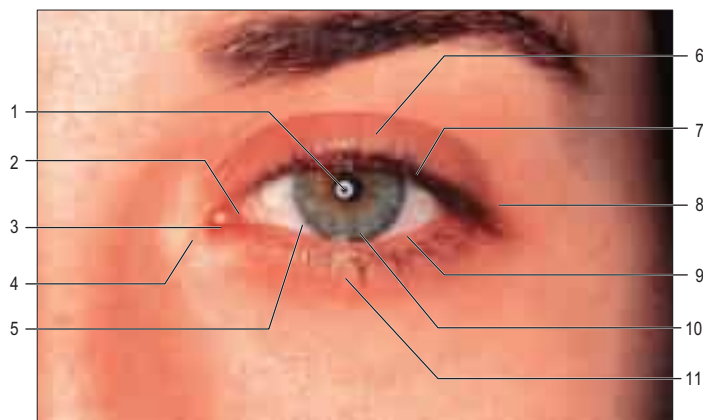


Fig. 41.19 The eyelids and anterior aspect of the eyeball. Key: 1, pupil; 2, plica semilunaris; 3, lacrimal caruncle; 4, medial canthus; 5, conjunctiva; 6, upper eyelid; 7, eyelashes; 8, lateral canthus; 9, lid margin; 10, iris; 11, lower eyelid. (With permission from Berkovitz BKB, Moxham BJ 2002 Head and Neck Anatomy. London: Martin Dunitz.)

the eyelashes and is termed the ciliary part of the eyelid. The margin medial to the papilla lacks eyelashes and forms the lacrimal part of the eyelid.

When looking straight ahead, the upper eyelid overlaps the upper part of the cornea by 2–3 mm, whereas the lower lid margin lies just below the corneoscleral junction (limbus). When the eyelids are closed, the upper lid moves down to cover the whole of the cornea. Malposition of the lower eyelid is common, particularly in the elderly. Ectropion describes the rolling out of the lower eyelid so that it is no longer in contact with the cornea, leading to epiphora (watering). Entropion describes the inversion of the eyelid with corresponding inturning of the eyelashes (trichiasis), which contact the cornea and cause irritation.

Each eyelid margin is 2–3 mm thick from front to back. The anterior two-thirds is skin and the posterior third is conjunctival mucosa. A narrow 'grey line' lies anterior to the mucocutaneous junction. This corresponds to the location of the ciliary (marginal) part of orbicularis oculi and is an important surgical landmark, since an incision at this point allows the eyelid to be split into anterior and posterior lamellae along a relatively bloodless plane. The eyelashes lie in front of the grey line, and the circular openings of the tarsal glands (Meibomian glands) lie behind it. The tarsal glands are often visible through the palpebral conjunctiva, when the eyelids are everted, as a series of parallel, faint yellow lines arranged perpendicular to the lid margins.

Eyelashes are short, thick, curved hairs, arranged in double or triple rows. The upper lashes, which are longer and more numerous, curve upwards, while those in the lower lid curve down, so that upper and lower lashes do not interlace when the lids are closed.

The eyelids pass to adjacent facial skin without obvious demarcation, although their limits are clearly defined in pathological conditions such as oedema. Various skin folds or furrows are of topographical interest. A prominent superior palpebral furrow or fold lies approximately opposite the upper margin of the tarsal plate and is deeply recessed when the lids are open. Asians have a skin flap, the epicanthus, which begins laterally in the superior palpebral fold and progresses medially to cover the medial canthus. Infant Caucasians frequently have a transient epicanthus, which sometimes persists in the adult. A less prominent inferior palpebral furrow occupies a similar position in the skin of the lower lid and deepens on downward gaze. A nasojugal furrow extends obliquely from the medial lower margin of the bony orbit to the cheek, and a malar furrow may be seen laterally along the inferior orbital rim in middle age, but only infrequently and faintly in the young.

Structure

From its anterior surface inwards, each eyelid consists of skin, subcutaneous connective tissue, fibres of the palpebral part of orbicularis oculi (see Fig. 30.18A), submuscular connective tissue, the tarsal plate (tarsus) with its tarsal glands and orbital septum, and palpebral conjunctiva (Fig. 41.20). The upper lid also contains the aponeurosis of levator palpebrae superioris.

The skin is extremely thin and is continuous at the palpebral margins with the conjunctiva. The subcutaneous connective tissue is very delicate, seldom contains any adipose tissue, and lacks elastic fibres.

The palpebral part of orbicularis oculi is subdivided anatomically into ciliary, pretarsal and preseptal parts. The palpebral fibre bundles are thin and pale, and lie parallel with the palpebral margins. Deep to them is the submuscular connective tissue, a loose fibrous layer that is continuous in the upper lid with the subaponeurotic layer of the scalp; effusions of blood or pus at this level can therefore pass down from the scalp into the upper eyelid. The main nerves lie in the submuscular layer, which means that local anaesthetics should be injected deep to orbicularis oculi.

Tarsal plates

The two tarsal plates (Fig. 41.21A) are thin, elongated, crescent-shaped plates of firm, dense fibrous tissue approximately 2.5 cm long. There is one in each eyelid to provide support and determine eyelid form. Each is convex and conforms to the configuration of the anterior surface of the eye. The free ciliary border is straight and adjacent to the eyelash follicles. The orbital border is convex and attached to the orbital septum. The superior tarsus, the larger of the two, is semi-oval, approximately 10 mm in height centrally. Its inferior edge is parallel to, and approximately 2 mm from, the lid margin. The smaller inferior tarsus is narrower and approximately 4 mm in vertical height.

The tarsal plates are connected to the margins of the orbit by the orbital septum and by the medial and lateral palpebral (canthal) ligaments (see Fig. 41.21A). The medial palpebral ligament passes from the medial ends of the two tarsal plates to the anterior lacrimal crest and

The palpebral fissure undergoes complex changes during infancy ([Paiva et al 2001](#)). At birth, the upper eyelid is at its lowest position and the margin of the lower eyelid is close to the centre of the pupil. Between the ages of 3 and 6 months, the upper eyelid reaches its maximum height and then declines in a linear fashion. The distance between the centre of the pupil and the margin of the lower eyelid increases linearly until the age of 18 months, when it stabilizes in position. The most common pattern of the lower eyelid crease is a single crease at birth and a double crease by the age of about 3 years. Ageing mainly affects the size of the horizontal eyelid fissure, which lengthens by about 10% between the ages of 12 and 25, and shortens by almost the same amount between middle age and old age. The peak level of growth in the horizontal dimension of the palpebral fissure is reached between the ages of 17 and 19 years, in the vertical dimension between 10 and 13 years, and in the intercanthal distance between 14 and 16 years in Asian children ([Park et al 2008](#)).

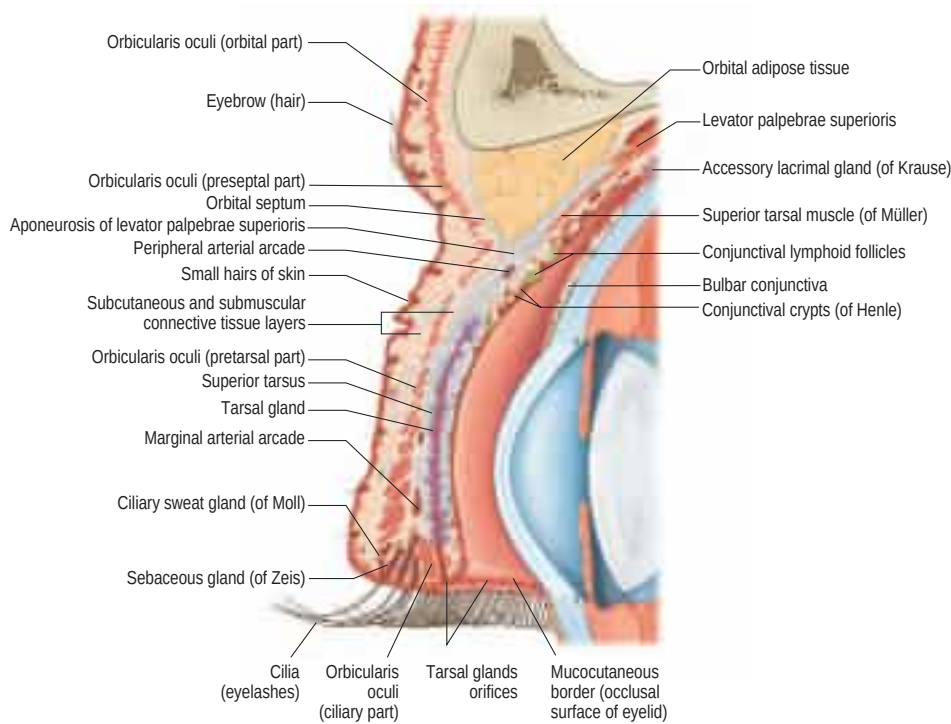


Fig. 41.20 The upper eyelid and anterior segment of the eye, sagittal section.

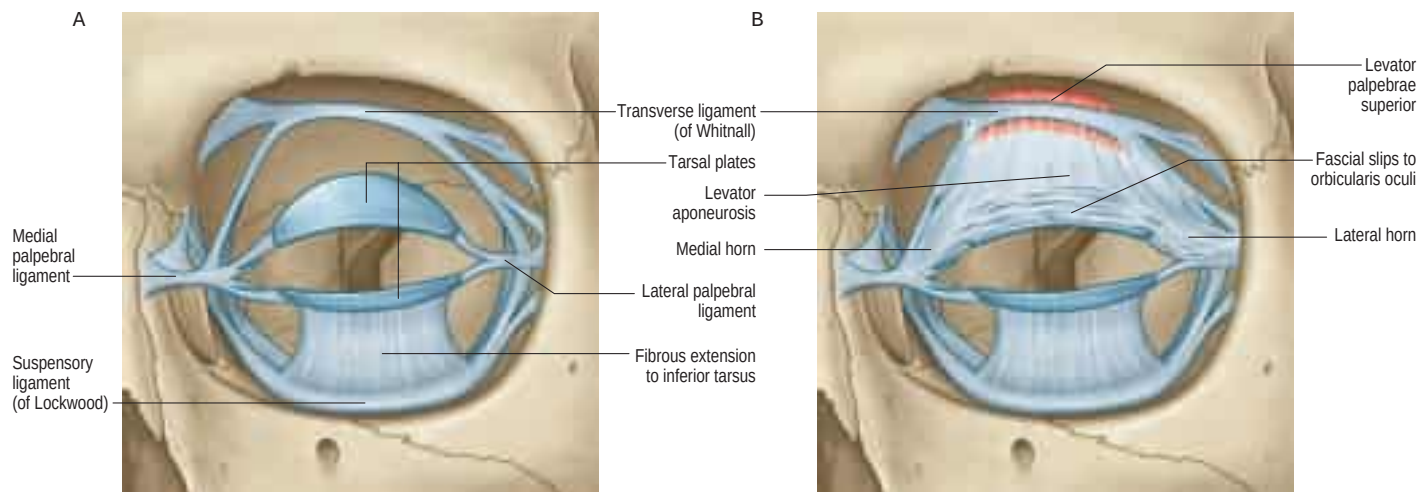


Fig. 41.21 **A**, The tarsal plates and their attachments. **B**, The anterior insertion of levator palpebrae superioris.

the frontal process of the maxilla. It splits at its insertion into the tarsal plates to surround the lacrimal canaliculi, and lies in front of the nasolacrimal sac and the orbital septum. The lateral palpebral ligament is relatively poorly developed. It passes from the lateral ends of the tarsal plates to a small tubercle on the zygomatic bone within the orbital margin and is more deeply situated than the medial palpebral ligament. It lies beneath the orbital septum and the lateral palpebral raphe of orbicularis oculi.

The deepest fibres of the aponeurosis of levator palpebrae superioris are attached to the anterior surface of the superior tarsus. The superior and inferior tarsal plates are also associated with a thin lamina of smooth muscle forming the superior and inferior tarsal muscles, respectively. Opposite the equator of the eye, the superior tarsal muscle passes from the inferior face of levator palpebrae superioris to a fibrous extension that projects to the upper margin of the superior tarsus (see Fig. 41.20). The muscle is innervated by the sympathetic nervous system and, on contraction, elevates the eyelid. Although it may be regarded as supplementing the action of the levator muscle, its full role is not clear. The mild ptosis that is a characteristic feature of Horner's syndrome is the result of an interruption to the sympathetic supply to the superior tarsal muscle. A corresponding but less prominent inferior tarsal muscle in the lower eyelid unites the inferior border, and possibly also the anterior surface, of the inferior tarsus to the capsulopalpebral fascia, which is the anterior expansion of the fused fascial sheath of inferior rectus and inferior oblique. Contraction of inferior rectus during downward gaze

therefore also pulls the lower lid downwards. The lower lid is capable of depressing by 4–5 mm, although it is not equipped with a striated muscle counterpart to the levator of the upper lid.

Palpebral glands

Tarsal (Meibomian) glands are modified sebaceous glands embedded in the tarsi. They are yellow and arranged in approximately 25 parallel rows perpendicular to the eyelid margin in the upper lid, and slightly fewer in the lower lid. They occupy the full tarsal height and are therefore longer centrally where the tarsi are higher. Each gland consists of a straight tube with many lateral secretory diverticula, and opens by a minute orifice on the free palpebral margin. It is enclosed by a basement membrane, and is lined at its orifice by stratified epithelium and elsewhere by a single layer of polyhedral cells. The sebaceous secretion of the tarsal glands spreads over the margins of the eyelids, and so an oily layer is drawn over the tear film as the palpebral fissure opens after a blink, reducing evaporation and contributing to tear film stability. The presence of the oily, hydrophobic secretions of tarsal glands along the margins of the eyelids also inhibits the spillage of tears on to the face. Obstruction of the tarsal gland ducts by lipid and cellular debris may result in lipogranulomatous inflammation and the clinical manifestations of an internal hordeolum or chalazion.

Small sebaceous glands (of Zeis) and sweat glands (of Moll) are associated with the eyelashes (see Fig. 41.20) and are prone to infection, producing an external hordeolum (stye).

Vascular supply and lymphatic drainage

The arterial supply to the eyelids is derived principally from the medial and lateral palpebral branches of the ophthalmic artery. Their branches course laterally along the tarsal edges to form superior and inferior arcades (two in the upper eyelid and one in the lower). The eyelids are also supplied by branches of the infraorbital, facial, transverse facial and superficial temporal arteries.

The veins that drain the eyelids are larger and more numerous than the arteries; they pass either superficially to veins on the face and forehead, or deeply to the ophthalmic veins within the orbit.

The lymph vessels that drain the eyelids and conjunctiva commence in a superficial plexus beneath the skin, and in a deep plexus in front of and behind the tarsi. These plexuses communicate with one another, and medial and lateral sets of vessels arise from them. The lateral lymph vessels drain the whole thickness of the lateral part of the upper and lower lids, and pass laterally from the lateral canthus to end in the superficial and deep parotid lymph nodes. The lymph vessels of the medial set drain the skin over the medial part of the upper eyelid, the whole thickness of the medial half of the lower lid, and the caruncle; they follow the course of the facial vein to end in the submandibular group of lymph nodes.

Innervation

The sensory innervation of the eyelids comes from both the ophthalmic and maxillary divisions of the trigeminal nerve. The upper eyelid is supplied mainly by the supraorbital branch of the frontal nerve. Additional contributions come from the lacrimal nerve, the supratrochlear branch of the frontal nerve, and the infratrochlear branch of the nasociliary nerve. The nerve supply to the lower eyelid is derived principally from the infraorbital branch of the maxillary nerve, with small contributions from the lacrimal and infratrochlear nerves.

CONJUNCTIVA

The conjunctiva is a thin, transparent mucous membrane that extends from the eyelid margins anteriorly, providing a lining to the lids, before turning sharply on itself to form the fornices; from here, it is reflected on to the globe to cover the sclera up to its junction with the cornea. The conjunctiva thus forms a sac that opens anteriorly through the palpebral fissure (Fig. 41.22). At the free palpebral margins, the conjunctiva is continuous with the skin of the eyelids, the lining epithelium of the ducts of the tarsal glands, and with the lacrimal canaliculi and lacrimal sac (see below). The continuity between the conjunctiva and the nasolacrimal duct and nasal mucosa is important in the spread of infection.

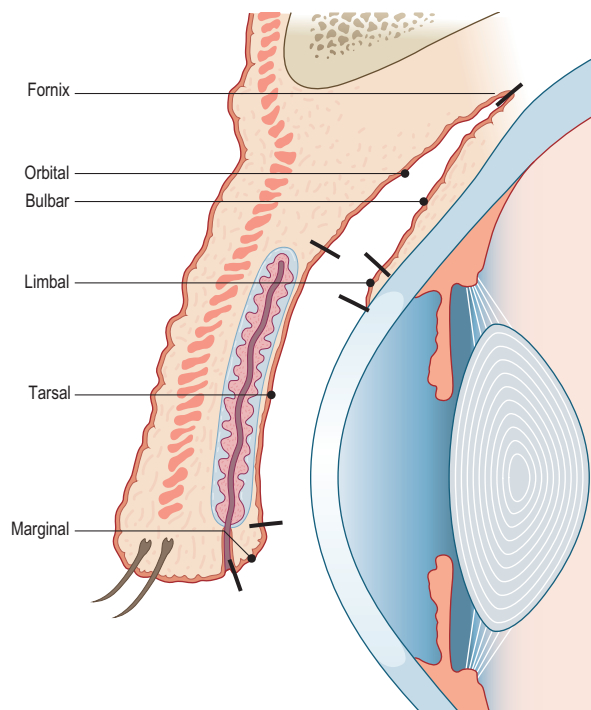


Fig. 41.22 The regions of the conjunctiva, sagittal section.

Conjunctival goblet cells contribute the mucin component of the preocular tear film and play a central role in the defence of the ocular surface against microbial infection. The conjunctiva is conventionally divided into five regions: marginal, tarsal, orbital, bulbar and limbal (see Fig. 41.22). The marginal, tarsal and orbital regions are collectively referred to as the palpebral conjunctiva.

The marginal zone extends from a line immediately posterior to the openings of the tarsal glands and passes around the eyelid margin to continue on the inner surface of the lid as far as the subtarsal groove (a shallow sulcus that marks the marginal border of the tarsus). The tarsal conjunctiva is highly vascular and is firmly attached to the underlying tarsal plate. The orbital zone extends as far as the fornices, which mark the line of reflection of the conjunctiva from the lids on to the eyeball. The conjunctiva is more loosely attached to underlying tissues over the orbital zone and so folds readily. Elevations of the conjunctival surface in the form of papillae and lymphoid follicles (part of the mucosa-associated lymphoid tissue, MALT) are commonly observed in this region (Knop and Knop 2002).

Ducts of the lacrimal gland open into the lateral part of the superior fornix. The bulbar conjunctiva is loosely connected to the eyeball over the exposed sclera, is thin and transparent, and readily permits the visualization of conjunctival and episcleral blood vessels. The loose attachment of the conjunctiva to the fascial sheath of the globe (Tenon's capsule) in this region means that the conjunctiva is freely movable here; as the bulbar conjunctiva approaches the cornea, its surface becomes smoother and its attachment to the sclera increases. The limbal conjunctiva extends approximately 1–1.5 mm around the cornea and contains a dense network of capillaries.

Structure

The conjunctiva is composed of an epithelial layer and an underlying fibrous layer or substantia propria. The form of the epithelium and thickness of the substantia propria vary with location. At the margin of the lids, the epithelium is non-keratinized stratified squamous and 10–12 cells thick. The epithelium of the tarsal conjunctiva thins to two or three layers and consists of columnar and flat surface cells. Near the fornices, the cells are taller, and a trilaminar conjunctival epithelium covers much of the bulbar conjunctiva. It thickens closer to the corneoscleral junction and then changes to stratified squamous epithelium typical of the cornea. A proportion of limbal conjunctival epithelial cells serve as stem cells for the corneal epithelium; this region of the conjunctiva is therefore essential for maintaining corneal integrity.

Mucus-secreting goblet cells are scattered within the conjunctival epithelium. They show a marked regional variation in density, being most frequent in conjunctival regions normally covered by the eyelids, and reduced in number within exposed interpalpebral locations (Doughty 2012).

The substantia propria, or fibrous layer, is thickest at the fornix and thinnest over the tarsi where conjunctival attachment is firmest. With the exception of the tarsal conjunctiva, the substantia propria adjacent to the epithelium is mainly loose. It merges with the fibrous fascia bulbi and episclera in the limbal and bulbar regions, and is loosely attached to the sheaths of the recti. At the fornix, it is continuous with orbital fascial tissues and has loose attachments to the overlying tendon of levator palpebrae superioris and the superior tarsal muscle fascia superiorly, an arrangement that provides support for the fornix during eyeball rotation.

Vascular supply and lymphatic drainage

The arteries of the palpebral conjunctiva are derived from the vascular arcades of the eyelids. Many of the small vessels of the eyelid drain to the conjunctival veins, which, in turn, pass back to the orbital and facial veins. The substantia propria of the tarsal and orbital conjunctiva is richly vascular. A dense capillary network is arranged in a single subepithelial lamina, a distribution that meets the requirement of the cornea during sleep, when oxygen is available to the cornea only through this vascular system. The bulbar conjunctiva receives blood from the palpebral arcades and from the anterior ciliary arteries that reach the conjunctiva on the orbital surface of the recti. The veins drain by the same route and join the ophthalmic veins.

Lymphatics drain into pre-auricular and submandibular lymph nodes.

Innervation

The conjunctiva is innervated by sensory nerves from the same ophthalmic and maxillary branches of the trigeminal nerve that serve the eyelid. Autonomic nerve fibres are abundant in the conjunctiva, particularly in bulbar and limbal regions, and are probably vasomotor in function.

LACRIMAL SYSTEM

The lacrimal system consists of structures responsible for the production of tears (principally, the main lacrimal gland with a contribution from accessory lacrimal glands) and the lacrimal drainage pathway that collects the tear fluid and conveys it into the nasal cavity (paired lacrimal canaliculi, lacrimal sac and nasolacrimal duct) (Fig. 41.23).

The total tear volume is approximately 7 μL . It is distributed within the precorneal tear film (1–2 μL), and along the upper and lower marginal tear strips (5–6 μL), which are wedge-shaped menisci that run along the posterior border of the lid margins and join together at the canthi.

Lacrimal gland

The lacrimal gland is the primary producer of the aqueous component of the tear layer. Its secretion is a watery fluid with an electrolyte content similar to that of plasma and containing several proteins that play key

roles in protecting the ocular surface from microbial infection (e.g. lysozyme, lactoferrin, immunoglobulin A (IgA)) and also maintaining ocular surface epithelial integrity.

The lacrimal gland consists of orbital and palpebral parts that are continuous posterolaterally around the concave lateral edge of the aponeurosis of levator palpebrae superioris. The orbital part, about the size and shape of an almond, lodges in a shallow fossa on the medial aspect of the zygomatic process of the frontal bone, just within the orbital margin. It lies above levator palpebrae superioris and, laterally, above lateral rectus. Its lower surface is connected to the sheath of levator palpebrae superioris and its upper surface is connected to the orbital periosteum. Its anterior border is in contact with the orbital septum and its posterior border attached to the orbital fat. The palpebral part, about one-third the size of the orbital part, is subdivided into two or three lobules and extends below the aponeurosis of levator palpebrae superioris into the lateral part of the upper lid, where it is attached to the superior conjunctival fornix. It is visible through the conjunctiva when the lid is everted. The lacrimal gland may be congenitally absent (Keith and Boldt 1986).

The main ducts of the lacrimal gland, up to 12 in number, discharge into the conjunctival sac at the superior lateral fornix. Those from the orbital part penetrate the aponeurosis of levator palpebrae superioris to join those from the palpebral part. Excision of the palpebral part is therefore functionally equivalent to total removal of the gland (Maitchouk et al 2000).

Many small accessory lacrimal glands (glands of Krause and Wolfring) occur in or near the fornix. They are more numerous in the upper eyelid, and their presence may explain why the ocular surface does not dry up after extirpation of the main lacrimal gland.

Microstructure

The lacrimal gland is lobulated and tubulo-acinar in form. Its secretory units are acini similar to those found in the salivary glands (Fig. 41.24) (Rusnell 1975). Acini consist of secretory cells that discharge their product into a central lumen continuous with an intercalated duct formed from a single layer of epithelial cells that lack secretory granules. Myoepithelial cells extend processes around the perimeter of acini and ducts; their contraction imparts a mechanical force on the acini and ducts, which promotes the expulsion of tears from the gland. The interstices of the gland are composed of loose connective tissue that contains numerous immune cells, mainly B-lymphocytes and plasma cells (particularly IgA-secreting cells).

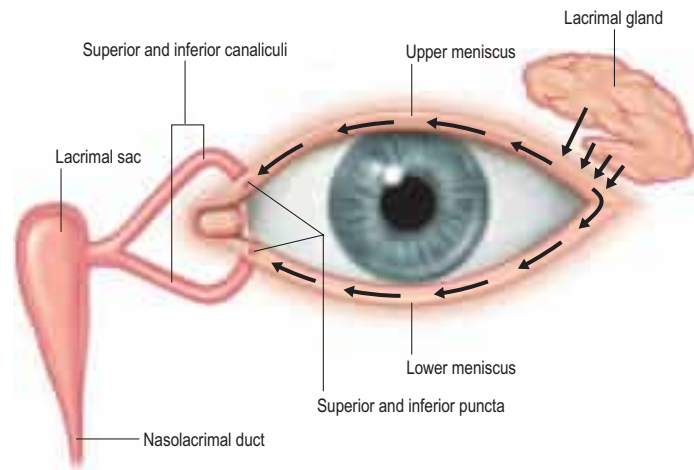


Fig. 41.23 A representation of tear dynamics on the ocular surface. Tears are produced by the lacrimal gland and flow within the upper and lower meniscus. Tears drain sequentially through the puncta, canaliculi, lacrimal sac and nasolacrimal duct.

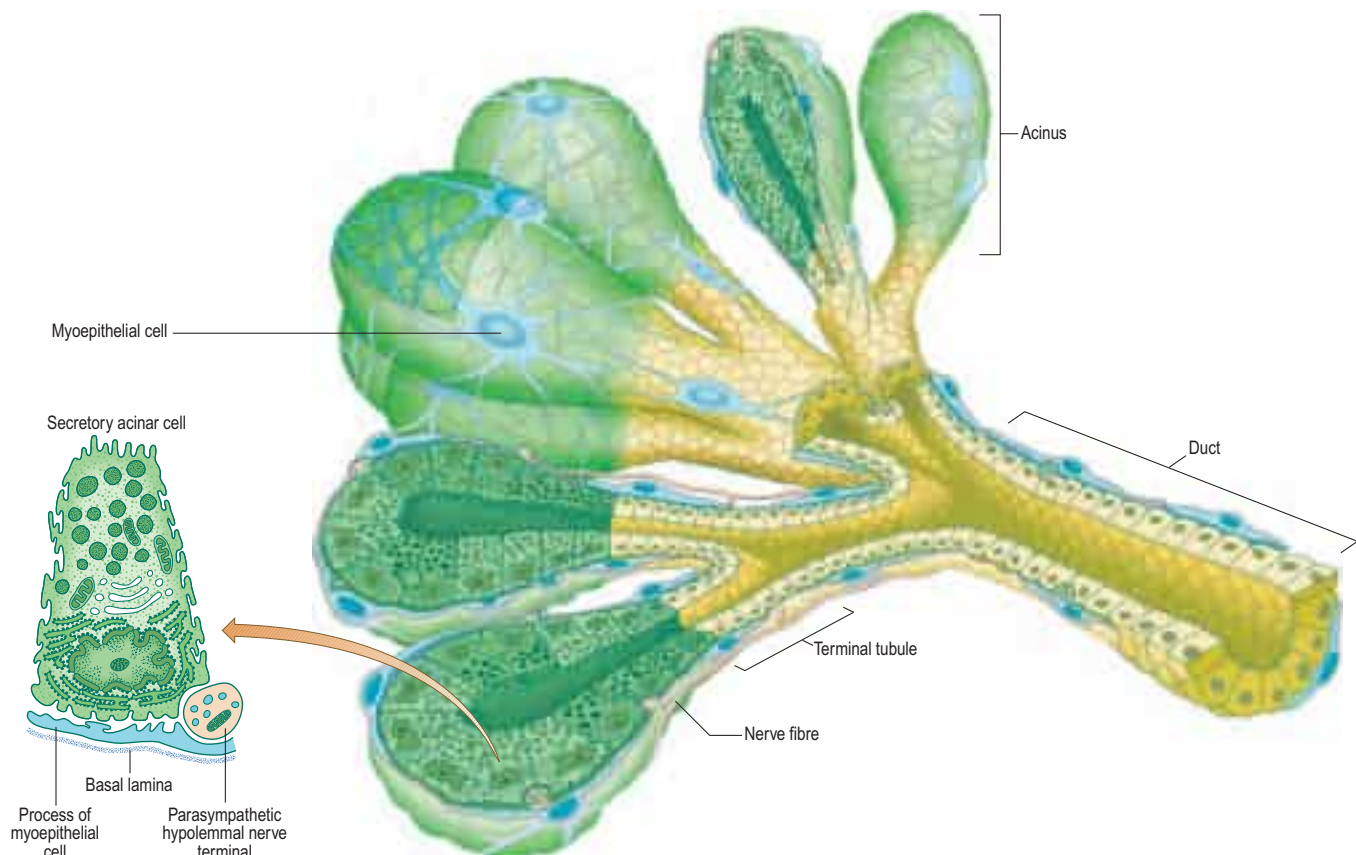


Fig. 41.24 The organization of the secretory units in the lacrimal gland.

Vascular supply

The lacrimal gland receives its arterial blood supply from the lacrimal branch of the ophthalmic artery. It may also receive blood from the infraorbital artery. Venous drainage is into the superior ophthalmic vein.

Innervation

The lacrimal gland is innervated by secretomotor postganglionic parasympathetic fibres from the pterygopalatine ganglion (Ch. 32). They reach the gland either via zygomatic and lacrimal branches of the maxillary nerve, or by passing directly from the ganglion (Ruskell 2004). Sympathetic fibres that issue from the superior cervical ganglion also supply the lacrimal gland. These fibres may be involved in the regulation of blood flow and the modulation of gland secretion (Dartt 2009).

Lacrimation reflex

The lacrimation reflex is stimulated by irritation of the conjunctiva and cornea, which elicits a large increase in tear volume (Situ and Simpson 2010). The afferent limb of the reflex involves branches of the ophthalmic nerve, with an additional contribution from the infraorbital nerve if the conjunctiva of the lower eyelid is involved. Impulses enter the brain and spread by interneurons to activate parasympathetic neurones in the superior salivatory nucleus and sympathetic neurones in the upper thoracic spinal cord. The efferent pathway to the lacrimal gland involves the greater petrosal nerve, which carries preganglionic parasympathetic secretomotor fibres, and the deep petrosal nerve, which conveys postganglionic sympathetic fibres; the parasympathetic fibres relay in the pterygopalatine ganglion, and the sympathetic fibres pass through the ganglion without synapsing.

Lacrimation may also occur in response to emotional triggers without any irritation of ocular structures, when it may be accompanied by alterations in the mimetic facial muscles, vocalizations and sobbing (Gračanin et al 2014).

Preocular tear film

The tear film is a complex fluid that covers the exposed parts of the ocular surface framed by the eyelid margins. Classically, the tear film has been regarded as a trilaminar structure, with a superficial lipid layer (secreted by the Meibomian glands) that overlies an aqueous phase (derived from the main and accessory lacrimal glands) and an inner mucinous layer (produced mainly by conjunctival goblet cells) (Fig. 41.25). The tear film performs a number of important functions. By smoothing out irregularities of the corneal epithelium, it creates an even surface of good optical quality that is reformed with each blink. The air-tear interface forms the principal refractive surface of the optical system of the eye. Since the cornea is avascular, it is dependent on the tear film for its oxygen provision. When the eye is open, the tear film is in a state of equilibrium with the oxygen in the atmosphere, and gaseous exchange takes place across the tear-epithelial interface. The constant turnover of the tear film also provides a mechanism for the removal of metabolic waste products. Tears play a major role in the defence of the eye against microbial colonization; the washing action of the tear fluid reduces the likelihood of microbial adhesion to the ocular surface, and the tears contain a host of protective antimicrobial proteins.

Lacrimal drainage pathway

There is a constant turnover of tears; production is matched by elimination. Some tears are lost by evaporation or absorption across the conjunctiva, but the majority are eliminated via the nasolacrimal drainage system (see Fig. 41.23). Tears collect at the medial canthal angle, where they drain into the puncta of the upper and lower lids, which are directed towards the surface of the eye to receive tear fluid. From each punctum, tears drain into lacrimal canaliculi. There is one canaliculus, approximately 10 mm long, in each lid. Each canaliculus first passes vertically from its punctum for about 2 mm and widens to form an ampulla, before passing medially towards the lacrimal sac. The superior canaliculus is smaller and shorter than the inferior. The canaliculi almost always unite to form a common canaliculus before reaching the lacrimal sac. Congenital absence of the lacrimal puncta, as well as supernumerary lacrimal puncta and canaliculi, have been described (Satchi and McNab 2010).

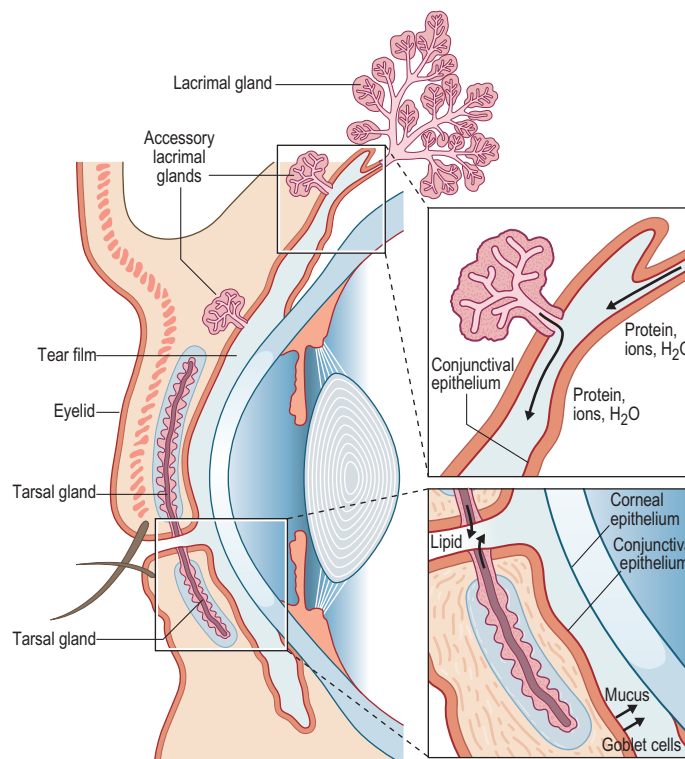


Fig. 41.25 Orbital glands that contribute the various components of the preocular tear film. The aqueous component (consisting primarily of proteins, ions and water) is produced by the main and accessory lacrimal glands. Tarsal glands in the eyelids produce the lipid layer of the tear film, and the mucous component is derived from conjunctival goblet cells.

The mucosa lining the canaliculi has a non-keratinized stratified squamous epithelium lying on a basement membrane, outside which is a lamina propria rich in elastic fibres (the canaliculi are therefore easily dilated when probed). Striated muscle fibres of orbicularis oculi interweave on each side of the canaliculus in a manner suggesting a sphincter-like spiralling, and supporting the claim that lumen size is regulated on blinking, possibly facilitating tear drainage.

The lacrimal sac is the closed upper end of the nasolacrimal duct. It is approximately 12 mm long and lies in a fossa in the lacrimal bone in the anterior part of the medial wall of the orbit (see Fig. 41.2). The sac is bounded in front by the anterior lacrimal crest of the maxilla and behind by the posterior lacrimal crest of the lacrimal bone. Its closed upper end is laterally flattened, its lower part is rounded and merges into the duct, and the lacrimal canaliculi open into its lateral wall near its upper end.

A layer of lacrimal fascia, continuous with the orbital periosteum, passes between the lacrimal crest of the maxilla and the lacrimal bone. It forms a roof and lateral wall to the lacrimal fossa and separates the lacrimal sac from the medial palpebral ligament in front and the lacrimal part of orbicularis oculi behind. A plexus of minute veins lies between the fascia and the sac. The upper half of the lacrimal fossa is related medially to the anterior ethmoidal sinuses, and the lower half to the anterior part of the middle meatus. The lacrimal sac has a fibro-elastic wall and is lined internally by mucosa that is continuous with the conjunctiva through the lacrimal canaliculi, and with the nasal mucosa through the nasolacrimal duct (Paulsen 2003).

The nasolacrimal duct is approximately 18 mm long, and descends from the lacrimal sac to open anteriorly in the inferior meatus of the nose at an expanded orifice. A fold of mucosa (plica lacrimalis) forms an imperfect valve just above its opening (ostium lacrimalis). The duct runs down an osseous canal formed by the maxilla, lacrimal bone and inferior nasal concha. It is narrowest in the middle and is directed downwards, backwards and a little laterally. The mucosa of the lacrimal sac and the nasolacrimal duct has a bilaminar columnar epithelium, which is ciliated in places. A rich plexus of veins forms erectile tissue around the duct; engorgement of these veins may obstruct the duct.

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